

A Quantitative Study of Technical Analysis Indicators and Intraday Trading Strategies on Indian Equity Market Data

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Abstract

This project evaluates whether common technical analysis indicators provide predictive signals in intraday equity markets and whether those signals translate into robust trading performance. Using 5-minute OHLC data for 49 NIFTY50 constituents from January to April 2026, indicators spanning trend (SMA/EMA, MACD), momentum (RSI, Stochastic, ROC), volatility (Bollinger Bands, ATR), and volume (OBV, VWAP, A/D) are computed and tested.

First, an event-based signal framework measures forward returns after indicator triggers across multiple horizons. Pooled results show that average effects are close to zero, with nominal significance disappearing after multiple-testing correction, indicating limited standalone predictive power. Second, the study designs and backtests single-asset and cross-sectional strategies, incorporating filtering and combination rules. Performance is mixed and highly regime-dependent, with transaction costs and drawdowns limiting practical viability. Overall, the findings suggest that isolated technical indicators are insufficient for intraday alpha, and that any usable edge likely depends on careful signal filtering, risk controls, and execution-aware strategy design.

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Chapter 1

Introduction

1.1 Motivation

Intraday markets now offer dense, low-latency data and practical access to algorithmic execution. This has made technical indicators a popular toolkit for short-horizon decision making, yet much of their usage remains heuristic. In liquid equities, small statistical edges are easily eroded by costs and microstructure noise, so indicator effectiveness must be demonstrated rather than assumed Fama 1970; Lo, Mamaysky, and Wang 2000.

1.2 Problem Statement

This report asks a focused question: do widely used technical indicators contain predictive information at 5-minute horizons, and can that information be converted into stable trading performance in the NIFTY50 universe? The investigation spans both signal-level predictability and strategy-level outcomes under realistic intraday constraints.

1.3 Objectives

- Quantify the predictive power of indicator signals using event-based forward returns.
- Compare the robustness of signals across assets and horizons.
- Design and backtest intraday strategies, including filters and combinations, to assess tradability.

1.4 Contributions

- A unified evaluation pipeline that links indicator signals to both statistical and trading performance.

- A pooled, cross-asset signal analysis with multiple-testing control for intraday indicators.
- Comparative evidence on single-asset and multi-asset strategy behavior in a realistic NIFTY50 setting.

1.5 Thesis Structure

Chapter 2 reviews technical analysis indicators and notation. Chapter 3 describes the NIFTY50 intraday dataset and preprocessing. Chapter 4 evaluates the predictive power of indicator signals. Chapters 5 and 6 present single-asset and cross-sectional backtesting results, including filtered and combined strategies. The final chapter summarizes findings and outlines future work.

Chapter 2

Background & Technical Analysis Indicators

2.1 Overview of Technical Analysis

Technical analysis studies market behavior through price and volume history, assuming that information is reflected in prices and that patterns can recur over time Murphy 1999. In this thesis, technical indicators are treated as quantitative summaries of trend, momentum, volatility, and volume that may carry predictive content Lo, Mamaysky, and Wang 2000. The goal of the background is to define the core indicators used later while keeping the presentation concise and consistent with established references Fama 1970; Murphy 1999.

2.2 Notation

Table 2.1 defines the symbols used in the formulas below.

2.3 Core Indicator Families

2.3.1 Trend

The Simple Moving Average (SMA) over window n is

$$SMA_t = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i} \quad (2.1)$$

and the Exponential Moving Average (EMA) is

$$EMA_t = \alpha P_t + (1 - \alpha) EMA_{t-1} \quad (2.2)$$

Table 2.1: Symbol definitions for technical indicator formulas

Symbol	Meaning
P_t	price at time t (close unless stated)
V_t	volume at time t
n	lookback window length
α	EMA smoothing factor, $2/(n + 1)$
SMA_t	simple moving average at time t
EMA_t	exponential moving average at time t
H_t, L_t	high and low at time t
C_{t-1}	prior close
r_t	return $(P_t - P_{t-1})/P_{t-1}$
σ_t	rolling standard deviation of P_t or r_t
k	band width multiplier (typically 2)
RS	average gain to average loss ratio
H_n, L_n	highest and lowest price over last n periods
TR_t	true range
OBV_t	on-balance volume
$VWAP_t$	volume-weighted average price
AD_t	accumulation/distribution line
MF_t	money flow multiplier

where $\alpha = 2/(n + 1)$. EMA reacts faster to recent price changes, while SMA provides smoother trends Murphy 1999. The MACD summarizes trend and momentum via the difference between fast and slow EMAs:

$$MACD_t = EMA_t^{(12)} - EMA_t^{(26)} \quad (2.3)$$

and is commonly used with a signal line for crossover interpretation Appel 1979.

2.3.2 Momentum

The Relative Strength Index (RSI) is

$$RSI = 100 - \frac{100}{1 + RS} \quad (2.4)$$

and flags overbought or oversold conditions Wilder 1978. The Stochastic Oscillator compares the close to its recent range,

$$K_t = \frac{P_t - L_n}{H_n - L_n} \times 100 \quad (2.5)$$

while the Rate of Change (ROC) measures percentage change over n periods,

$$ROC = \frac{P_t - P_{t-n}}{P_{t-n}} \times 100 \quad (2.6)$$

both of which summarize momentum Murphy 1999.

2.3.3 Volatility

Bollinger Bands are defined around an SMA using a multiple of standard deviation,

$$Upper = SMA_t + k\sigma_t, \quad Lower = SMA_t - k\sigma_t \quad (2.7)$$

and are widely used for volatility-aware signals Bollinger 2001. The Average True Range (ATR) uses the true range,

$$TR_t = \max\{H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|\} \quad (2.8)$$

to quantify typical price movement Wilder 1978. Rolling standard deviation itself is

$$\sigma_t = \sqrt{\frac{1}{n} \sum_{i=0}^{n-1} (P_{t-i} - \mu)^2} \quad (2.9)$$

and serves as a baseline volatility measure Murphy 1999.

2.3.4 Volume

On-balance volume is defined as

$$OBV_t = OBV_{t-1} + \begin{cases} V_t, & P_t > P_{t-1} \\ -V_t, & P_t < P_{t-1} \\ 0, & \text{otherwise} \end{cases} \quad (2.10)$$

and tracks cumulative volume flow Murphy 1999. VWAP is

$$VWAP_t = \frac{\sum P_t V_t}{\sum V_t} \quad (2.11)$$

and acts as an intraday benchmark for institutional execution Murphy 1999. The Accumulation/Distribution line is

$$AD_t = AD_{t-1} + MF_t \cdot V_t \quad (2.12)$$

capturing buying and selling pressure Murphy 1999.

2.4 Indicator Summary Table

Table 2.2 provides a compact overview of the indicators used in the study.

Table 2.2: Indicator families, categories, and primary use

Indicator	Category	Primary use
SMA, EMA, MACD	Trend	Direction and crossover signals
RSI, Stochastic, ROC	Momentum	Overbought/oversold and speed
Bollinger, ATR, σ_t	Volatility	Risk and regime context
OBV, VWAP, A/D	Volume	Participation and pressure

2.5 Why These Indicators?

These indicators are widely used, computationally lightweight, and interpretable, which makes them suitable for intraday analysis where noise, costs, and timing matter Murphy 1999. The subsequent chapters evaluate their predictive usefulness and how filtering or combinations alter performance in practice Lo, Mamaysky, and Wang 2000.

Chapter 3

Data Description and Preprocessing

3.1 Data Source and Collection

Market data is obtained from Yahoo Finance through an automated pipeline. The dataset consists of 5-minute intraday price data over approximately 60 trading days (January 2026 to April 2026), with all timestamps aligned to Indian Standard Time (IST).

The asset universe is based on the NIFTY 50 index. Data was successfully retrieved for 49 equities, while one constituent (Tata Motors) was excluded due to unavailability. A validation step ensures that only datasets with sufficient observations and consistent structure are retained.

A summary of dataset coverage is shown in Figure 3.1.

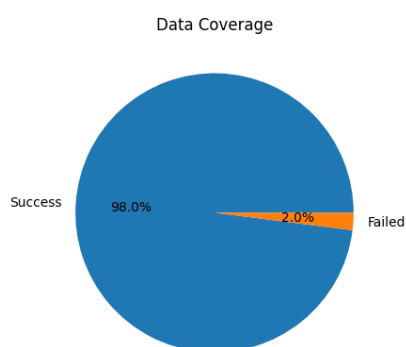


Figure 3.1: Proportion of successfully retrieved and missing assets from the NIFTY 50 universe

Each dataset contains standard OHLC (Open, High, Low, Close) price fields, which form the basis for all subsequent analysis.

3.2 Data Preprocessing

To ensure consistency, a preprocessing pipeline is applied to each time series. Observations are first sorted chronologically to maintain a proper time index. Duplicate timestamps are removed, and missing values are handled using forward filling to preserve continuity.

These steps produce a clean and aligned dataset suitable for time-series analysis and indicator computation.

3.3 Intraday Structure

The dataset reflects the structure of trading sessions in the Indian equity market. Each trading day consists of sequential 5-minute intervals covering active market hours.

Figure 3.2 illustrates a representative intraday price series.

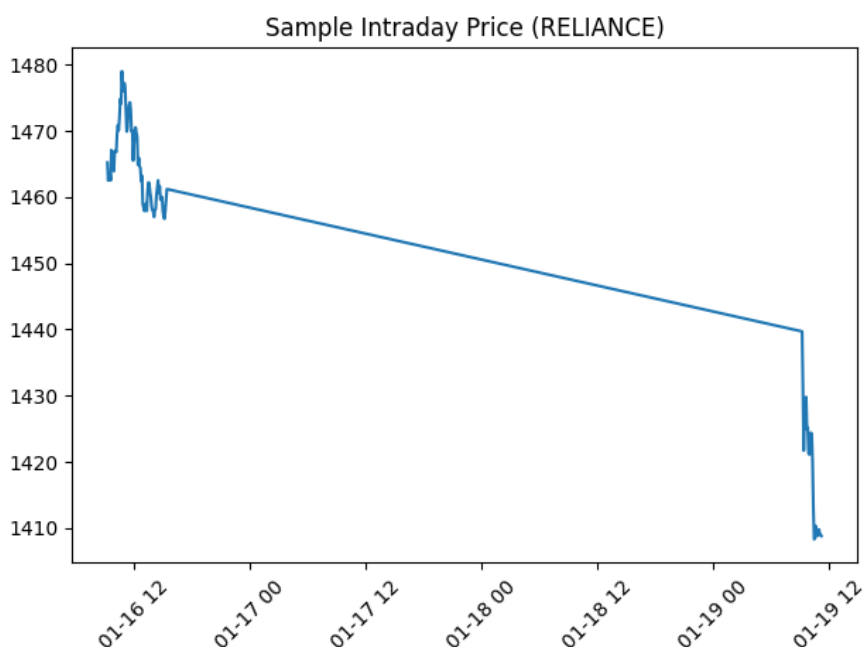


Figure 3.2: Sample intraday price series at 5-minute resolution

The dataset excludes non-trading hours, ensuring that all observations correspond to active market periods.

3.4 Data Representation

The processed data is organized as a time-indexed series, where each row represents a 5-minute interval and columns correspond to OHLC prices.

This structure enables efficient computation of rolling statistics and technical indicators while maintaining proper temporal alignment.

3.5 Dataset Overview

Figure 3.3 shows the sector-wise distribution of the selected equities, indicating broad coverage across industries.

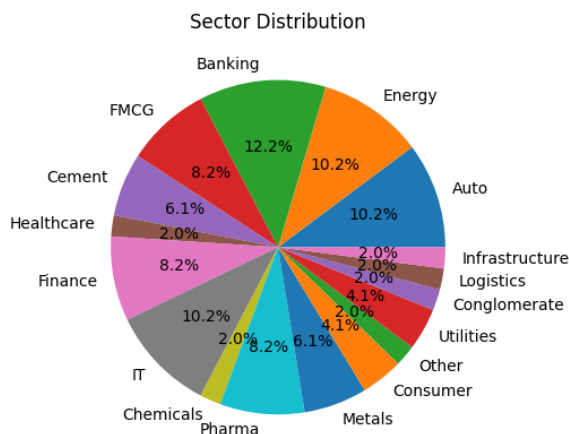


Figure 3.3: Sector-wise distribution of selected equities

Key dataset characteristics are summarized in Table 3.1.

Table 3.1: Dataset overview and summary characteristics

Metric	Value
Total Assets Considered	50
Successfully Retrieved	49
Failed Retrievals	1
Sampling Frequency	5-minute intervals
Observation Window	~60 trading days
Time Period	Jan 2026 – Apr 2026
Price Fields	Open, High, Low, Close

The resulting dataset provides a consistent foundation for evaluating technical indicators in an intraday setting.

Chapter 4

Signal Analysis: Predictive Power of Technical Indicators

4.1 Introduction and Objective

Given a signal generated by an indicator, does it have predictive power over future returns?

Technical indicators are widely used to infer market behavior, yet their effectiveness should be validated statistically rather than assumed. This chapter evaluates the *predictive power* of indicator-based signals across the NIFTY50 universe, focusing on whether they exhibit statistically and economically meaningful relationships with subsequent returns. The framing is grounded in the technical analysis literature Wilder 1978; Lo, Mamaysky, and Wang 2000 and tested against market efficiency Fama and French 1991; Fama 1970.

The analysis focuses on directional consistency, horizon persistence, and cross-asset stability.

4.2 Methodology

4.2.1 Overview of the Approach

The analysis uses an event-based framework: signals are identified per stock, forward returns are computed after each signal, and results are pooled across assets. This isolates signal information without simulating trades and allows market-level inference.

4.2.2 Signal Generation and Event Definition

For each stock, technical indicators are computed using 5-minute intraday price data. Signals are generated based on either threshold conditions (e.g., RSI crossing a level) or structural changes (e.g., moving average crossovers).

A key feature of the methodology is the use of *event-based signals*. Instead of recording a signal at every time step where a condition holds, a signal is recorded only at the moment the condition transitions from false to true. This ensures that each signal corresponds to a distinct market event and prevents repeated counting during persistent conditions.

4.2.3 Forward Return Computation

Once signal events are identified, predictive content is evaluated through forward returns. Let P_t denote the closing price at the time of a signal. The forward return over a horizon of h bars is defined as:

$$r_{t,h} = \frac{P_{t+h} - P_t}{P_t} \quad (4.1)$$

This measures the percentage price change following the signal.

Multiple horizons ($h \in \{1, 5, 10\}$) capture immediate and short-term effects.

4.2.4 Pooling Across Assets and Events

For each signal and horizon, forward returns are computed for all signal occurrences across all stocks and then combined into a single pooled dataset.

Pooling treats all signal instances as draws from a common distribution, increasing precision and allowing a market-level answer to whether a signal works on average. This is reasonable for large-cap index constituents with comparable microstructure.

4.2.5 Statistical Evaluation

Predictive performance is assessed using descriptive and inferential statistics. Descriptive metrics include mean return, standard deviation, and win rate (fraction of positive returns).

To formally test for predictability, a one-sample t -test is conducted under the null hypothesis:

$$H_0 : \mathbb{E}[r_{t,h}] = 0$$

This evaluates whether the average forward return differs significantly from zero.

Since 24 signal-horizon combinations are tested, multiple hypothesis testing is addressed using the False Discovery Rate (FDR) correction via the Benjamini-Hochberg procedure Benjamini and Hochberg 1995.

4.2.6 Data and Parameter Configuration

The dataset consists of 5-minute intraday price data for 49 NIFTY50 stocks over roughly 60 trading days (about 4,350 observations per stock). Forward returns are evaluated over

$h \in \{1, 5, 10\}$ bars (5, 25, and 50 minutes), yielding 3,500–8,300 observations per signal-horizon combination.

4.2.7 Technical Indicators

The analysis considers four widely-used indicators representing distinct trading philosophies: trend-following (SMA, EMA), momentum/mean-reversion (RSI), and volatility-based signals (Bollinger Bands) Wilder 1978; Lo, Mamaysky, and Wang 2000; Bollinger 2001.

Moving average crossover signals use short- and long-term windows (5 vs 15 for SMA, 10 vs 30 for EMA). RSI uses a 14-bar window with 30/70 thresholds Wilder 1978. Bollinger Bands use a 20-bar moving average with ± 2 standard deviation bands Bollinger 2001. Together, these indicators provide a representative cross-section of common technical strategies.

4.3 Pooled Empirical Results and Analysis

This section presents the pooled evaluation and interprets statistical and economic significance.

4.3.1 Summary of Predictive Performance

Table 4.1 shows that average returns are extremely small (about -5.2×10^{-4} to $+2.1 \times 10^{-4}$), and win rates remain near the 50% benchmark. Only a few signal-horizon combinations show nominal significance, and none survive FDR correction, consistent with random variation. No systematic improvement appears across horizons.

Table 4.1: Pooled signal results with FDR-corrected significance

Signal	h	N	Mean	Win %	t	p (FDR)
SMA_bull	1	8301	-5.443900626129467e-06	0.47162992410552945	-0.2401680678912945	0.8454322419970681
SMA_bull	5	8299	-7.951327525694843e-05	0.48319074587299676	-1.5015358113069284	0.31981219410705003
SMA_bull	10	8290	-2.455030715358335e-05	0.4854041013268999	-0.31439444128452637	0.8419024126567781
SMA_bear	1	8272	6.218243413001836e-05	0.4993955512572534	2.5148937296716705	0.057243053741543024
SMA_bear	5	8258	0.00010887602684148176	0.5060547348026156	2.0224001506536413	0.14800184566499924
SMA_bear	10	8250	-3.865899446235052e-05	0.4932121212121212	-0.4898346679046798	0.8419024126567781
EMA_bull	1	3584	2.551144775818455e-05	0.4779575892857143	0.5973551164313594	0.8419024126567781
EMA_bull	5	3580	3.599685102434366e-05	0.48743016759776536	0.40754854951065506	0.8419024126567781
EMA_bull	10	3572	0.00014152090213514376	0.5081187010078387	1.1333186047511234	0.5610688643756278
EMA_bear	1	3579	0.00010279867226211719	0.508521933500978	2.5738805179177047	0.057243053741543024
EMA_bear	5	3573	2.868991477630755e-05	0.5088161209068011	0.3048471828439906	0.8419024126567781
EMA_bear	10	3572	-0.0002518439078781673	0.4946808510638298	-1.8959160129164832	0.1741554249456932
RSI_bull	1	7011	3.2185087585306694e-05	0.4946512623020967	0.991082263647956	0.6433592407462684
RSI_bull	5	7010	-5.196046033199374e-05	0.4978601997146933	-0.791150795018447	0.7456161896412252
RSI_bull	10	7009	-0.0002812713195361223	0.4880867456127836	-3.03891914753454	0.02859646609295384
RSI_bear	1	6885	8.356287586652149e-06	0.4781408859840232	0.29010600702058903	0.8419024126567781
RSI_bear	5	6877	1.9411093866309613e-05	0.4725897920604915	0.3023134861168911	0.8419024126567781
RSI_bear	10	6869	-8.558392587352636e-06	0.4766341534430048	-0.09897302286079937	0.9211625780044935
BB_bull	1	5836	6.806480922677755e-05	0.504112405757368	1.641618311195579	0.2685947235783281
BB_bull	5	5833	-6.199176771633695e-05	0.50814332247557	-0.7808153821543726	0.7456161896412252
BB_bull	10	5831	-0.0005198617768512423	0.4860229806208198	-4.581460714442965	0.00011312545838552286
BB_bear	1	5858	1.3516104352466342e-05	0.47507681802663027	0.4143325392204417	0.8419024126567781
BB_bear	5	5855	0.00018994786955058523	0.48539709649871904	2.704374904101439	0.05490473939472499
BB_bear	10	5840	0.00021428586817507896	0.48476027397260274	2.0356219576621344	0.14800184566499924

4.3.2 Indicator-Level Observations

Across SMA/EMA crossovers, RSI, and Bollinger Bands, results are broadly consistent: effects are weak, unstable across horizons, and economically negligible. Deviations from zero are inconsistent with the intended signal interpretation.

4.3.3 Statistical vs. Economic Significance

The large sample size provides high statistical power, so even tiny deviations can appear statistically detectable. However, observed returns are orders of magnitude smaller than typical transaction costs, implying no practical tradability. Statistical detectability does not imply trading viability.

4.3.4 Visual Evidence

Figures 4.1–4.6 reinforce these findings: mean returns cluster near zero, win rates remain close to 50%, t-statistics are centered near zero, and return distributions are broadly symmetric. Together, these patterns confirm the absence of systematic predictive structure.

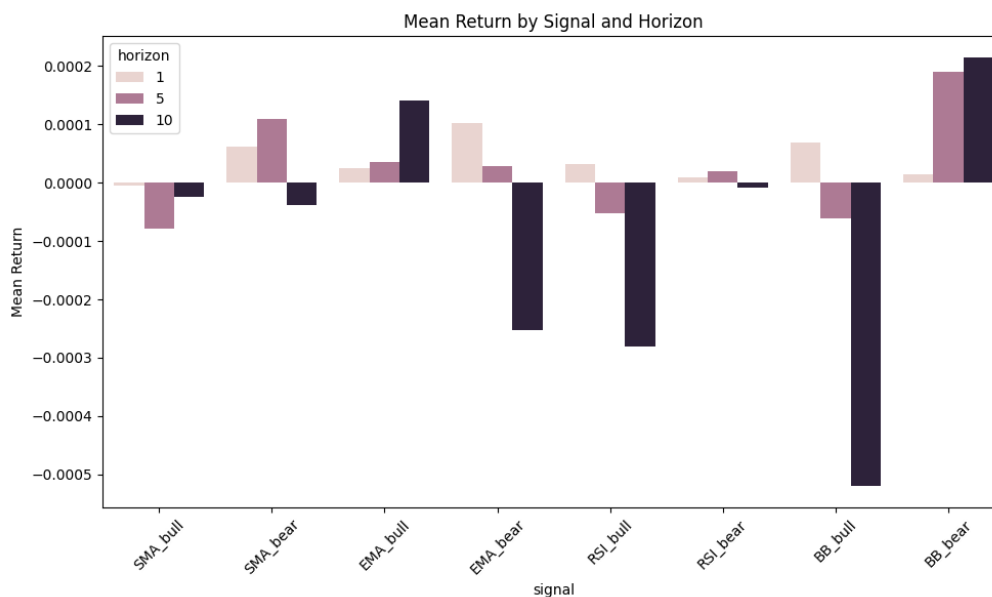


Figure 4.1: Mean returns by signal and horizon.

4.3.5 Implications for Signal Design and Market Behavior

The results suggest that commonly used technical indicators, when applied in isolation, do not provide actionable predictive signals in liquid intraday markets. This is consistent with the semi-strong form of market efficiency, where publicly available information is already reflected in prices Fama and French 1991. For practical trading, single-indicator signals are

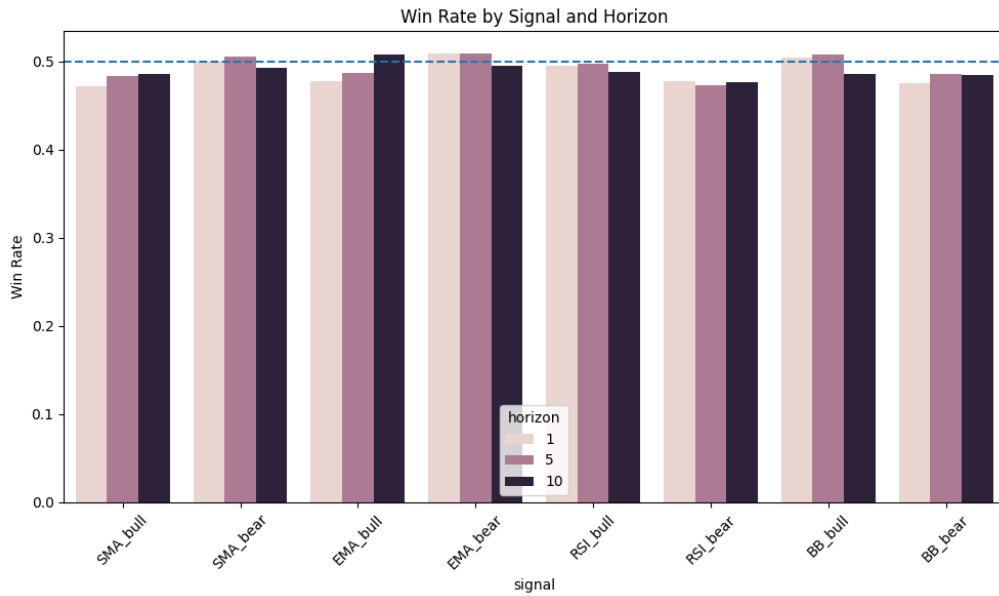


Figure 4.2: Win rates by signal and horizon.

insufficient; combining indicators, adding regime filters, and emphasizing execution and risk management are more likely to matter than any single signal.

4.3.6 Conclusion

Using a pooled dataset across 49 NIFTY50 stocks, this analysis finds no evidence that standard indicators (SMA, EMA, RSI, Bollinger Bands) deliver robust or economically meaningful predictive power in 5-minute data. Minor deviations from randomness are inconsistent, not robust to multiple testing, and too small to overcome costs. This motivates the shift to integrated strategy design and systematic backtesting in later chapters.

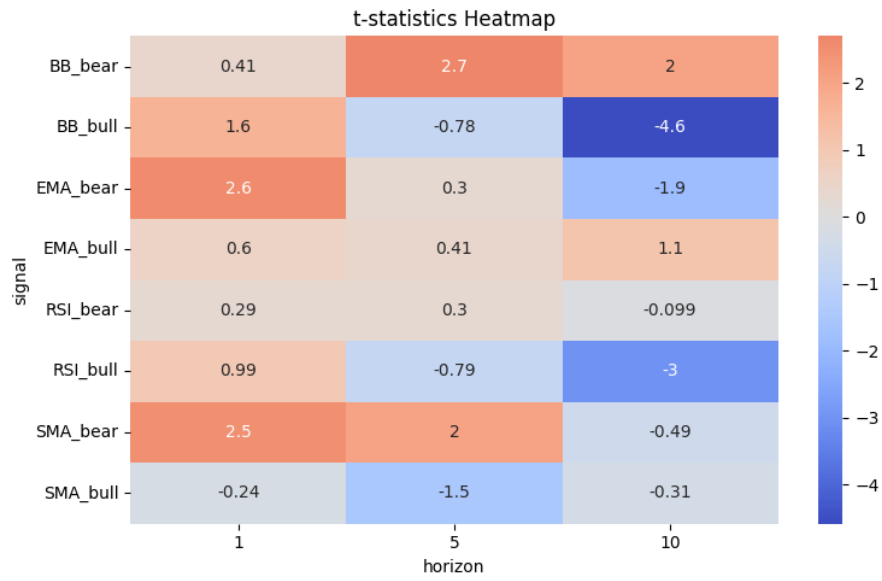


Figure 4.3: t-statistics heatmap across signals and horizons.

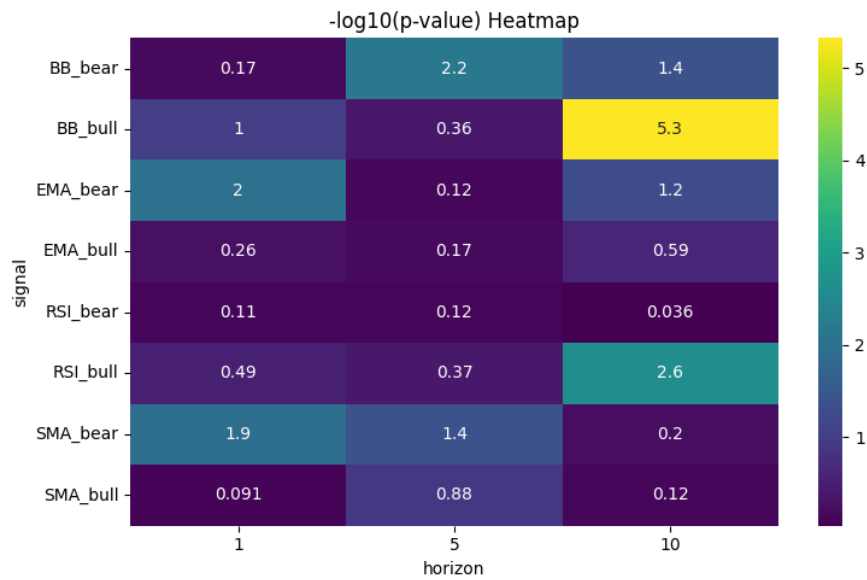


Figure 4.4: FDR-adjusted p -value heatmap (darker indicates lower p).

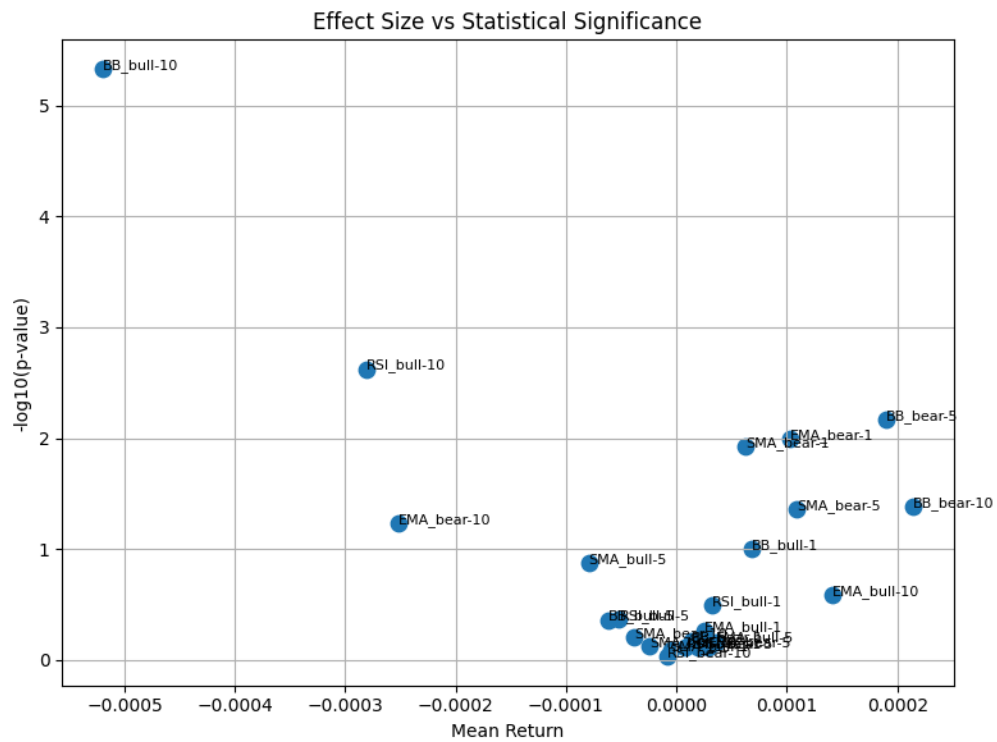


Figure 4.5: Effect size versus statistical significance.

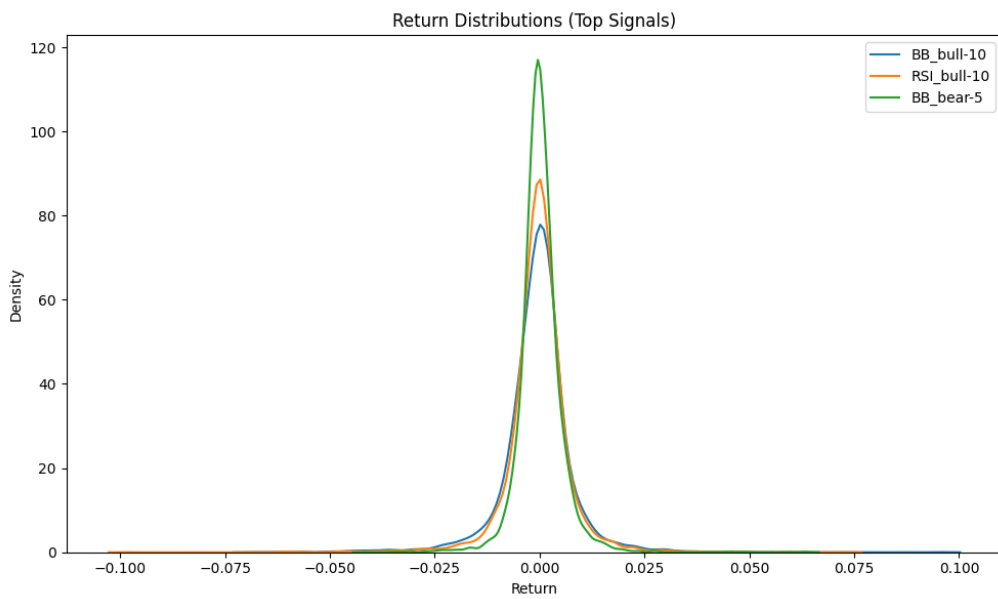


Figure 4.6: Return distribution following signals.

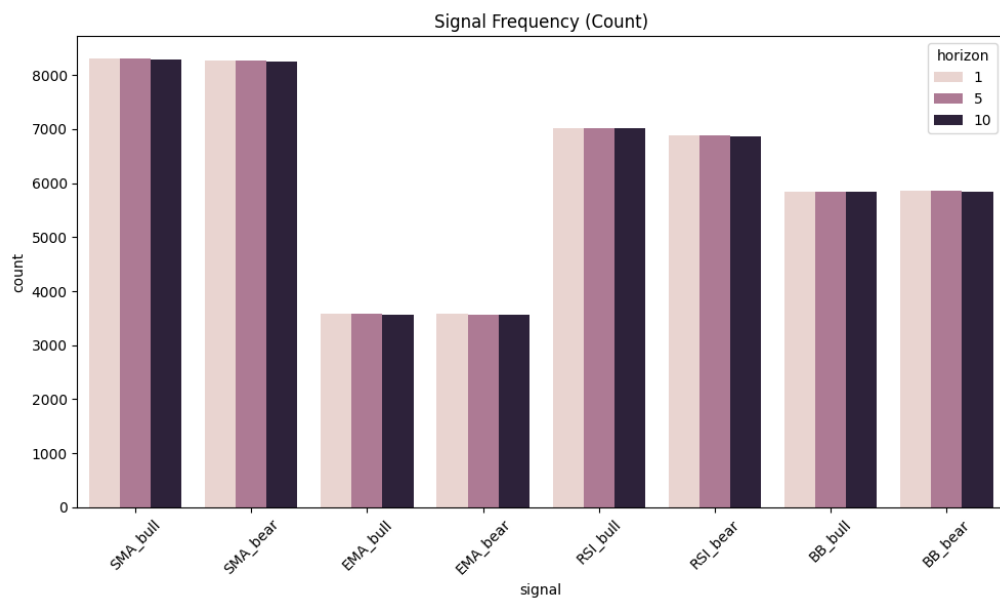


Figure 4.7: Signal counts by indicator and horizon.

Chapter 5

Single-Asset Backtesting and Strategy Evaluation

5.1 Introduction and Objective

Which intraday strategies produce robust, cost-aware performance on 5-minute data?

This chapter presents a systematic evaluation of intraday trading strategies on multiple NSE instruments using 5-minute OHLC data. The primary objective is to assess the effectiveness of baseline technical strategies under realistic trading conditions, explicitly accounting for transaction costs and execution frictions.

The study further investigates whether incorporating simple filters and combining multiple signals can improve performance robustness. Results are analyzed at the individual asset level and then aggregated across assets to identify consistent patterns and generalizable insights.

5.2 Data and Preprocessing

5.2.1 Data Source and Coverage

Data are sourced from Yahoo Finance via the `vectorbt` library Polakow 2021 and stored as 5-minute OHLC candles. The sample spans approximately 60 trading days in the Asia/Kolkata timezone, restricted to regular NSE market hours (09:15 to 15:30). The single-asset universe used in this experiment includes RELIANCE.NS, HDFCBANK.NS, ICICIBANK.NS, ^NSEBANK, and ^NSEI.

Each dataset follows the structure:

Datetime, Open, High, Low, Close

5.2.2 Preprocessing Pipeline

To ensure a consistent and reliable intraday structure, timestamps are sorted and duplicates removed, then the data are restricted to regular market hours. Missing 5-minute candles are detected on a per-day basis and logged for diagnostics, small gaps are forward-filled with a two-interval limit, and any remaining missing values are removed.

5.3 Strategy Design

5.3.1 Baseline Strategies

Baseline strategies are designed to represent common intraday trading paradigms. **Random** uses low-probability random entries and exits as a naive benchmark. **SMA Crossover** uses fast and slow simple moving average pairs (SMA_10_30 and SMA_5_20) with entry on a bullish crossover and exit on the reverse. **RSI Mean Reversion** uses a 14-period RSI with entries on oversold recoveries (30 or 25) and exits near mid-levels (55 or 60). **VWAP Mean Reversion** enters when price is sufficiently below the intraday VWAP and exits on reversion Algo 2023. **Opening Range Breakout (ORB)** enters on a breakout above the first 30-minute high and exits at the close Sahi 2023.

5.3.2 Combined Strategies

To improve robustness, baseline strategies are extended using simple filters that reduce noise and false signals. **ORB_VWAP** takes ORB breakouts only when price is above the intraday VWAP, using VWAP as directional confirmation. **RSI_TREND** conditions RSI mean-reversion signals on a trend filter that requires price to be above a 50-period moving average. **SMA_VOL** adds a volatility filter to SMA crossovers, allowing trades only when 20-period return volatility exceeds a predefined threshold.

All strategies operate on long-only signals and are executed within an intraday framework with forced end-of-day exits.

5.4 Backtesting Framework

The backtesting system is implemented using vectorbt Polakow 2021 and is designed as an event-driven simulation of intraday trading on 5-minute OHLC data. The objective is to evaluate strategy signals under realistic execution constraints while maintaining consistency across all tested models.

5.4.1 Signal-to-Trade Transformation

Each strategy generates binary entry and exit signals at each timestamp. To avoid repeated triggering under persistent conditions, signals are converted into discrete events. Let E_t and X_t denote raw entry and exit signals. Event signals are defined as:

$$E_t^{event} = E_t \cdot (1 - E_{t-1}), \quad X_t^{event} = X_t \cdot (1 - X_{t-1}) \quad (5.1)$$

This ensures that a position is opened only once per signal occurrence.

5.4.2 Execution Assumptions

All trades are executed under the following assumptions:

- Execution price: next candle open
- Position type: long-only
- Capital usage: full allocation per trade
- Holding period: intraday only (no overnight positions)

Positions are forcibly closed at the end of each trading session, ensuring zero overnight exposure and strict intraday consistency.

5.4.3 Transaction Costs and Risk Controls

Transaction costs are explicitly modeled to reflect real trading frictions. Each trade incurs:

- Fee: 0.05% per transaction
- Slippage: 0.02% per transaction

Risk is controlled using a fixed stop-loss mechanism:

$$SL = 1\% \quad (5.2)$$

No take-profit mechanism is used to avoid introducing additional optimization bias and to keep evaluation focused on signal quality rather than exit tuning.

5.4.4 Performance Evaluation

For each strategy and asset, the portfolio is constructed and evaluated using standard performance metrics:

- Total return

- Sharpe and Sortino ratios
- Maximum drawdown
- Win rate and profit factor
- Trade count and expectancy
- Return per trade

These metrics capture both risk-adjusted and raw performance, as well as trade efficiency.

5.4.5 Backtesting Algorithm

The overall procedure can be summarized as follows:

Algorithm 1 Intraday Event-Driven Backtesting Framework

```

1: for each asset do
2:   Load 5-minute OHLC data
3:   for each strategy do
4:     Compute entry and exit signals
5:     Convert signals to event triggers
6:     Simulate trades at next open price
7:     Apply transaction costs and slippage
8:     Enforce 1% stop-loss and end-of-day exit
9:     Compute portfolio statistics
10:  end for
11:  Rank strategies using return, Sharpe, drawdown score
12: end for
13: Aggregate results across assets for global evaluation

```

5.4.6 Evaluation Structure

The evaluation is performed in two stages. First, each asset is analyzed independently to obtain strategy-level rankings based on a composite score combining return, Sharpe ratio, and drawdown.

Second, results are aggregated across all assets to evaluate robustness. This cross-sectional view is used to assess whether performance is consistent or driven by specific market conditions.

Global analysis is visualized using average return comparisons, risk-return scatter plots, and heatmaps of strategy performance across assets.

5.5 Evaluation Metrics and Ranking

Each strategy is evaluated using a comprehensive set of performance, risk, and trade-level metrics derived from the simulated portfolio. Performance metrics include total return, Sharpe

ratio, Sortino ratio, and maximum drawdown. Trade-level diagnostics include win rate, profit factor, total trades, expectancy, average holding time, and total fees.

Return per trade normalizes performance by trading frequency and is computed as:

$$\text{return per trade} = \frac{\text{total return}}{\max(\text{total trades}, 1)}$$

Together, these measures separate signal quality (expectancy), trading frequency (number of trades), and cost efficiency (fees impact).

5.5.1 Per-Asset Ranking Methodology

For each individual asset, strategies are ranked using three core criteria: total return (higher is better), Sharpe ratio (higher is better), and maximum drawdown (lower is better). Each strategy s is assigned a rank for each metric: $r_s^{(ret)}$ for return (descending), $r_s^{(sharpe)}$ for Sharpe (descending), and $r_s^{(dd)}$ for drawdown (ascending).

The final composite score is defined as:

$$\text{score}_s = \frac{1}{3} (r_s^{(ret)} + r_s^{(sharpe)} + r_s^{(dd)})$$

Strategies are then sorted in ascending order of score_s , where a lower score indicates better overall performance.

5.5.2 Interpretation of Ranking

This ranking framework balances return maximization with risk control. High-return but high-drawdown strategies are penalized, low-risk but low-return strategies are also penalized, and strategies that achieve consistent risk-adjusted performance rank higher.

5.5.3 Cross-Asset Evaluation

Beyond per-asset rankings, robustness is evaluated by aggregating results across all tickers. Mean return, Sharpe, drawdown, and trade counts are computed per strategy, and cross-sectional consistency is assessed through average performance, the distribution of returns across assets, and the fraction of profitable instances. This ensures that top-ranked strategies are not driven by isolated performance on a single asset, but exhibit stability across different market conditions.

5.6 Global Aggregation and Cross-Asset Results

Per-asset results are aggregated into a global table that averages return, Sharpe, drawdown, and trade counts across assets. This provides a cross-asset view of robustness.

Table 5.1: Global average performance by strategy (all assets)

Strategy	Return	Sharpe	Drawdown	Trades
ORB_VWAP	0.021606940372899668	2.558439278540443	4.89304434945871	34.8
ORB_30	0.02051723386735085	2.438025259882627	4.989901132090521	35.0
SMA_VOL	0.006313360338625965	0.29607900396977305	2.9046770990144326	9.8
RSI_TREND	-0.012864425723423834	-5.222904994450243	1.8891978311192474	13.6
SMA_10_30	-0.03970529296480731	-4.747483969788844	9.193833506020608	79.4
Random	-0.04232005961486467	-9.552048034480082	4.750783244369537	19.0
SMA_5_20	-0.05187710399664776	-5.010114871620677	8.988102365180364	128.2
RSI_25_60	-0.08124054827557274	-11.707052528972445	8.988709288818216	59.6
RSI_30_55	-0.0973915649223685	-14.408487883651052	10.526605209306577	76.0
VWAP_0.005	-0.13092606450739547	-21.961474687162063	13.245697289877153	30.2

5.7 Single-Asset Results (Representative Case: RELIANCE_NS)

This section illustrates the per-asset outputs using RELIANCE_NS as a representative case. Each asset directory contains strategy-specific statistics and diagnostic plots.

5.7.1 Performance Summary

Table 5.2: RELIANCE_NS strategy summary (selected metrics)

Strategy	Return	Sharpe	Max DD	Trades
Random	-0.0030610862552248	-0.5404802108654997	2.8999950399377568	19
RSI_TREND	-0.005604090049547085	-2.6371611384504003	1.266338845835809	9
ORB_VWAP	-0.02228908460781633	-1.9450599665365935	5.958680585355518	35
ORB_30	-0.02228908460781633	-1.9450599665365935	5.958680585355518	35
SMA_VOL	-0.03016815185208252	-4.719167958049942	3.926372167503224	17
RSI_25_60	-0.060315813590193286	-7.0708611371196435	7.934828156860356	62
SMA_5_20	-0.06125278489493911	-5.439933516723417	6.721901878660471	128
SMA_10_30	-0.08526486713958249	-8.445927597516675	8.774957553154755	83
RSI_30_55	-0.09330646038863852	-12.123967486988171	10.682637628646553	76
VWAP_0.005	-0.18500008387659123	-26.18071711015435	18.62087972445198	42

5.7.2 Equity, Drawdown, and Trade Distribution

5.7.3 ORB_VWAP Example

5.8 Key Findings

Most standalone strategies are negative after costs, with high turnover amplifying fees. ORB is the most consistent baseline, suggesting early-session breakouts carry the strongest signal. VWAP reversion alone underperforms, but VWAP acts as an effective confirmation filter.

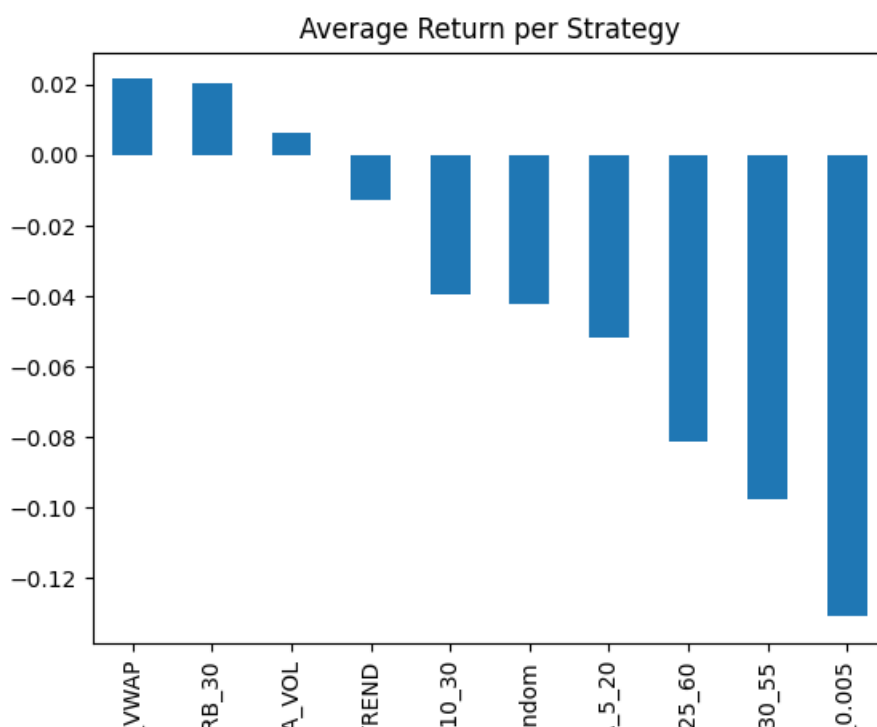


Figure 5.1: Average return by strategy across assets.

Combining strategies (ORB_VWAP, SMA_VOL, RSI_TREND) reduces noise trades and improves risk-adjusted results, while performance still varies by asset, making global averages essential to avoid overfitting to a single ticker.

5.9 Limitations and Future Work

The sample is short (about 60 trading days) and may not capture regime changes. Latency, partial fills, and order book effects are not modeled, position sizing is fixed to full allocation per trade, and the asset universe is small and concentrated in large-cap instruments. No regime detection or volatility targeting is applied. Future extensions include regime filters, dynamic position sizing, cross-sectional portfolio construction, and machine learning for signal conditioning.

5.10 Conclusion

Single-asset intraday backtests show that most simple technical indicators fail to overcome transaction costs at a 5-minute horizon. The Opening Range Breakout is the strongest baseline, and filtering it with VWAP produces the most robust average performance. These results motivate more selective trading and multi-asset portfolio construction in the next stage of the project.

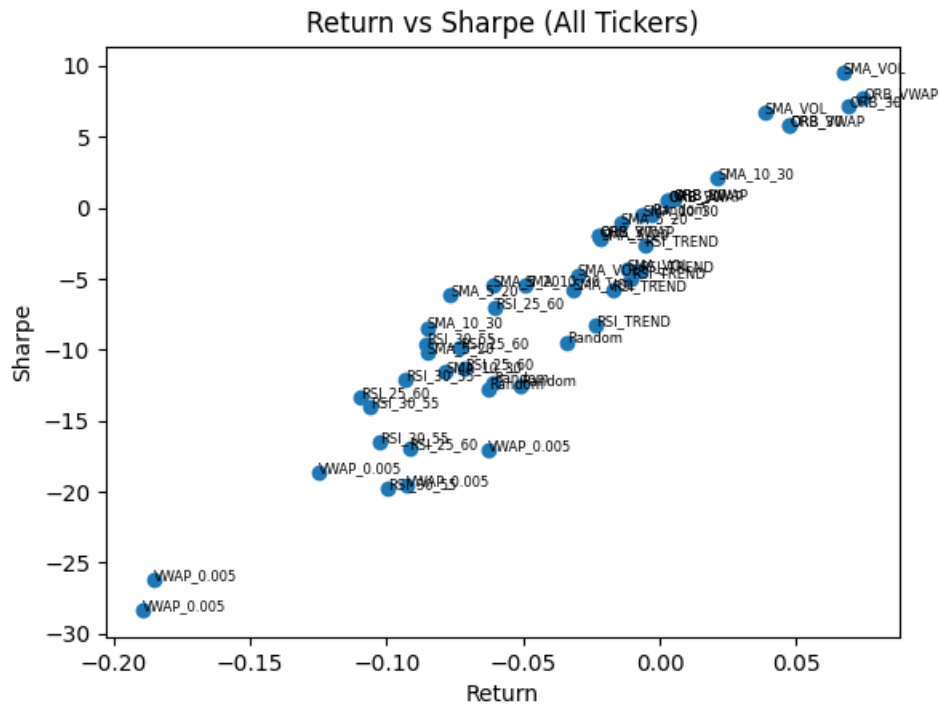


Figure 5.2: Risk-return scatter plot (return vs Sharpe) across all assets and strategies.

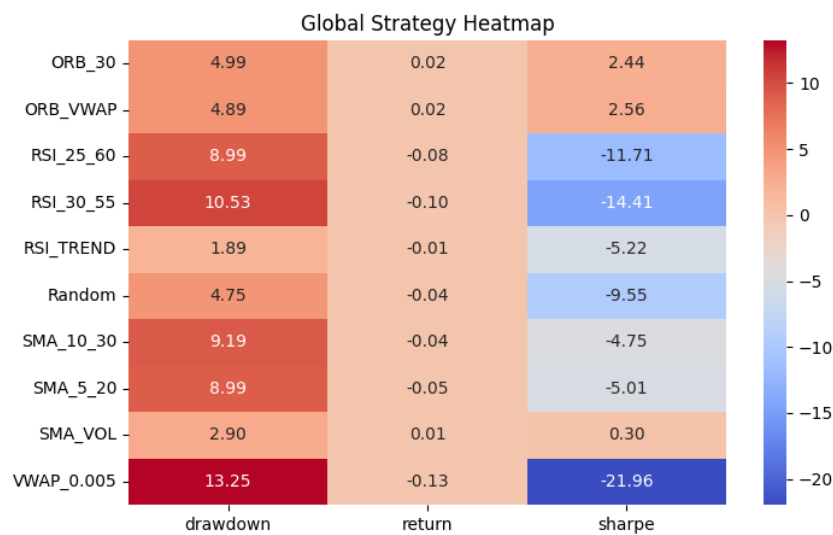


Figure 5.3: Global heatmap of average return, Sharpe, and drawdown by strategy.

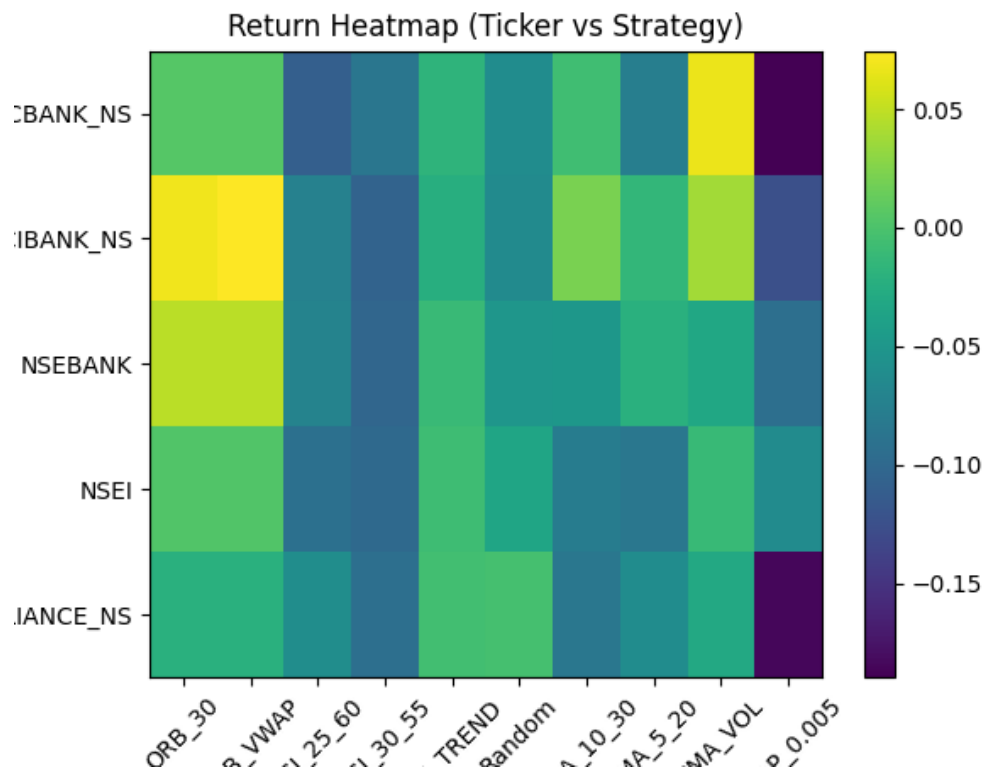


Figure 5.4: Return heatmap by asset and strategy.

Table 5.3: RELIANCE_NS ranking by composite score

Strategy	Final score
Random	1.3333333333333333
RSI_TREND	2.3333333333333335
ORB_VWAP	3.5
ORB_30	3.5
SMA_VOL	4.333333333333333
SMA_5_20	6.333333333333333
RSI_25_60	6.666666666666667
SMA_10_30	8.0
RSI_30_55	9.0
VWAP_0.005	10.0

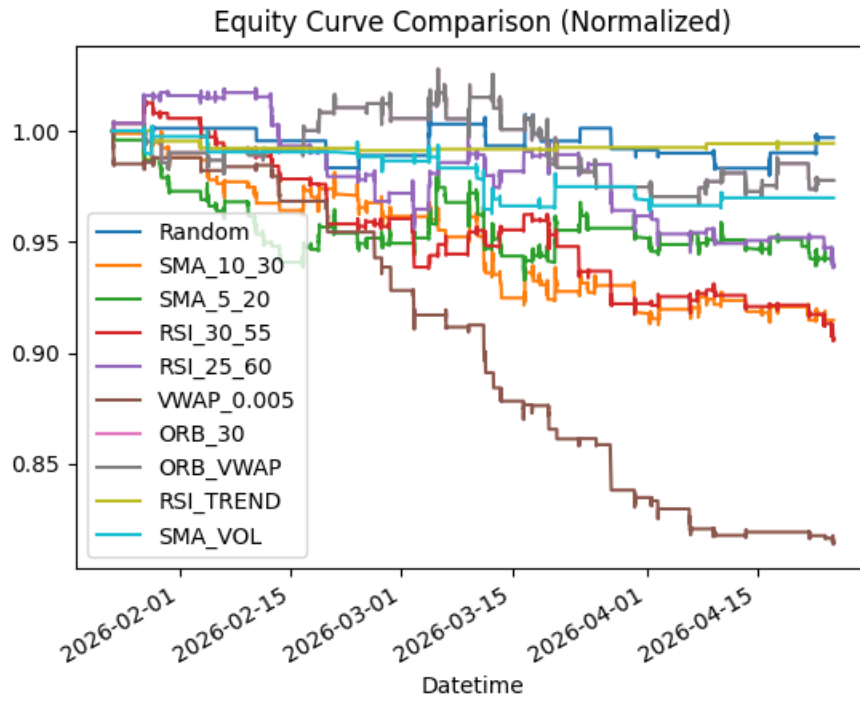


Figure 5.5: Normalized equity curves for RELIANCE_NS across strategies.

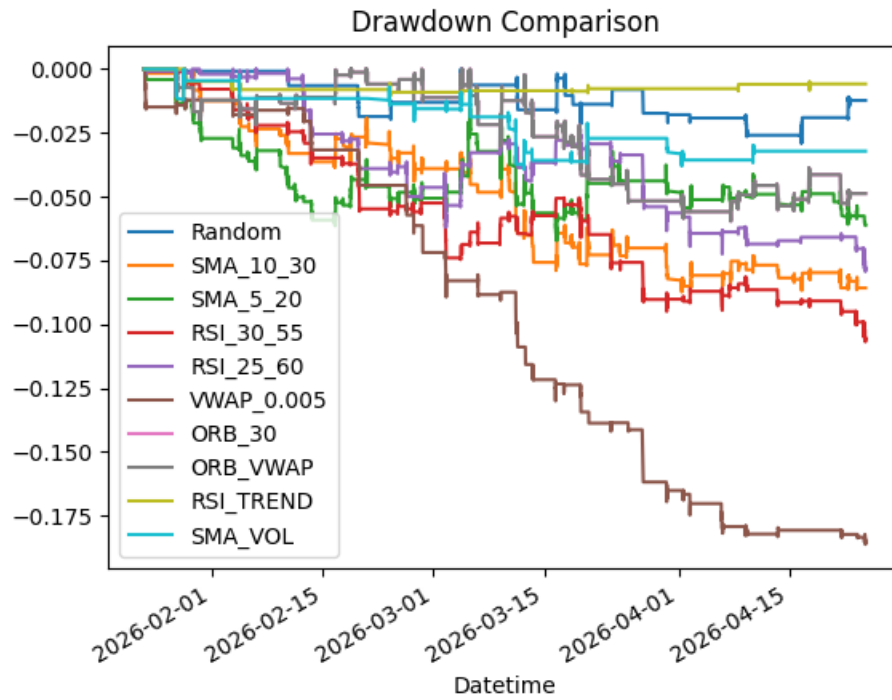


Figure 5.6: Drawdown comparison for RELIANCE_NS across strategies.

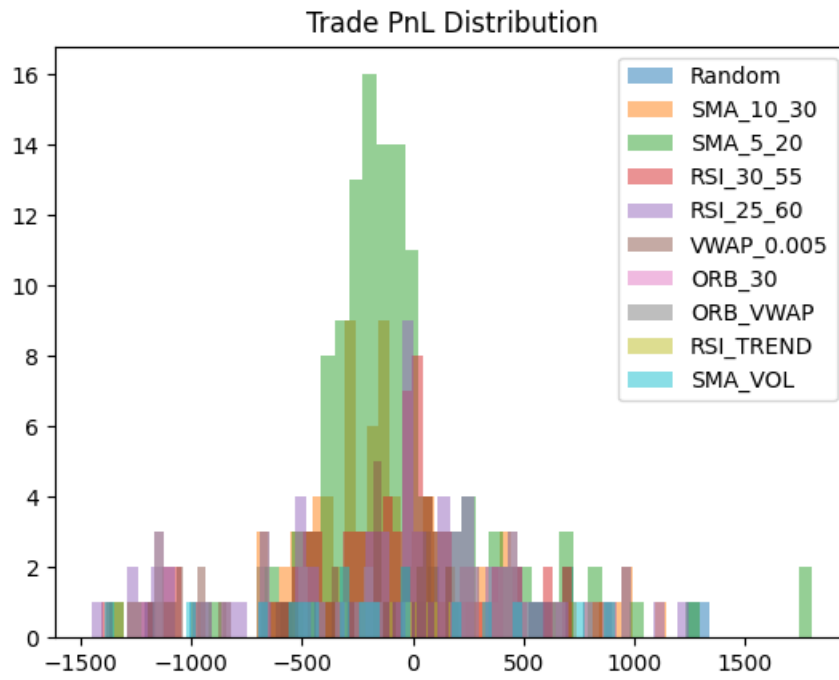


Figure 5.7: Trade PnL distributions for RELIANCE_NS.

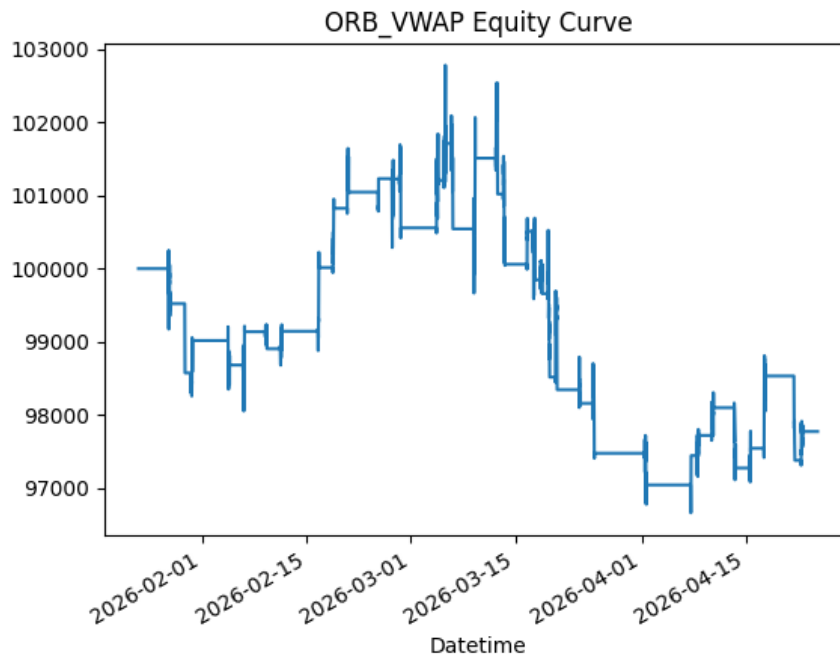


Figure 5.8: ORB_VWAP equity curve for RELIANCE_NS.

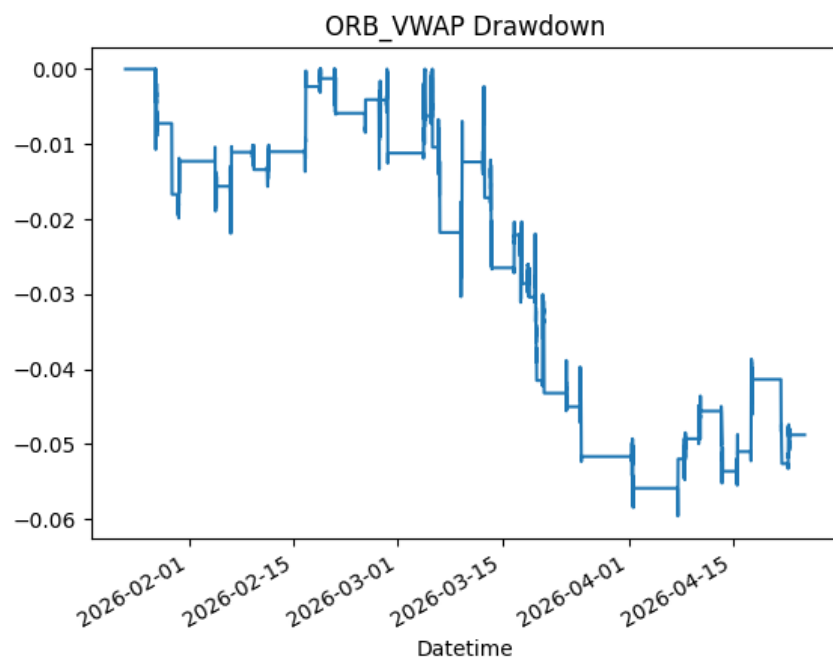


Figure 5.9: ORB_VWAP drawdown for RELIANCE_NS.

Chapter 6

Cross-Sectional Backtesting and Multi-Asset Robustness Analysis

6.1 Introduction

The preceding single-asset backtests showed a consistent pattern: several indicators exhibit non-random signal behavior, but this predictability does not translate into economically meaningful returns once transaction costs and execution frictions are incorporated.

Building on that foundation, this chapter extends the analysis to a cross-sectional, multi-asset framework. The objective is to test whether the single-stock behavior generalizes across the NIFTY50 universe or is driven by asset-specific idiosyncrasies.

Each stock is treated as an independent testbed and identical strategies are applied uniformly across assets. Rather than constructing a portfolio, outcomes are aggregated across asset-level backtests to assess signal robustness. This design isolates intrinsic signal quality while avoiding confounding effects from capital allocation or diversification mechanics.

The analysis proceeds through four layers of strategy design. First, pure indicator strategies evaluate standalone signal value. Second, filtered strategies condition signals on trend or volatility regimes to reduce noise. Third, combined strategies require multiple indicators to align, testing whether confirmation improves trade quality. Finally, stock-level analysis checks whether performance is consistent across assets or concentrated in a small subset.

Across all layers, a consistent backtesting framework is employed, and performance is evaluated using trade-level metrics that directly reflect the distribution of realized outcomes. This structured progression enables a systematic assessment of whether increasing model complexity leads to economically meaningful improvements in a cross-sectional setting, or whether fundamental limitations prevent technical indicator-based strategies from achieving robust profitability across assets.

6.2 Backtesting Framework

This section formalizes the process of converting indicator signals into executable trades in a multi-asset setting. The framework ensures consistent execution rules, avoids look-ahead bias, and enables comparable evaluation across assets.

6.2.1 Signal-to-Trade Conversion

Let $S_t \in \{0, 1\}$ denote a binary signal at time t , where $S_t = 1$ indicates a trade entry condition. To avoid look-ahead bias, all signals are executed with a one-bar delay:

$$\text{Execution time} = t + 1 \quad \text{for a signal generated at } t \quad (6.1)$$

This ensures that only information available up to time t is used in decision-making. Signal-to-trade conversion is implemented using `vectorbt` Polakow 2021, which maps entry and exit arrays into discrete trade events while preserving temporal consistency.

6.2.2 Trade Construction and Lifecycle

Each trade represents a complete round-trip position consisting of entry, holding, and exit. An exit occurs either through an opposing signal or predefined risk constraints.

For trade i , the net return is defined as:

$$R_i = \frac{P_{\text{exit},i} - P_{\text{entry},i}}{P_{\text{entry},i}} - c \quad (6.2)$$

where c denotes transaction costs.

Within each asset, trades are non-overlapping, ensuring at most one active position at any time. This constraint isolates signal quality by preventing interactions between simultaneous positions.

6.2.3 Execution Model and Assumptions

The framework uses 5-minute intraday data over approximately 60 trading days. Execution is modeled under the following assumptions:

- Execution at next bar open
- Long-only positions
- Full capital allocation per trade
- No overnight positions (end-of-day forced exit)

Transaction costs are fixed at a total of 0.07% per trade (fee plus slippage). A uniform initial capital of 100,000 is used only for equity curve computation and does not imply cross-asset capital allocation.

Each asset is evaluated independently under identical rules, ensuring that results reflect signal behavior rather than portfolio construction effects.

For combined strategies, deterministic risk controls are applied:

- Stop-loss: 1%
- Take-profit: 2%

These parameters are fixed ex-ante across all assets.

6.2.4 Performance Measurement

Performance is evaluated at the trade level. Let $\{R_1, R_2, \dots, R_N\}$ denote the set of trade returns.

Expected return is defined as:

$$\mathbb{E}[R] = \frac{1}{N} \sum_{i=1}^N R_i \quad (6.3)$$

This can be decomposed into winning and losing trades:

$$\mathbb{E}[R] = p \cdot \bar{R}_{\text{win}} + (1 - p) \cdot \bar{R}_{\text{loss}} \quad (6.4)$$

where p is the win probability.

The profit factor is defined as:

$$\text{PF} = \frac{\sum_{R_i > 0} R_i}{|\sum_{R_i < 0} R_i|} \quad (6.5)$$

Risk-adjusted performance is measured using:

$$\text{Sharpe} = \frac{\mathbb{E}[R]}{\sigma(R)} \quad (6.6)$$

Sharpe 1966

Maximum drawdown is computed from the equity curve:

$$\text{MDD} = \max_t \left(\frac{\text{Peak}_t - \text{Value}_t}{\text{Peak}_t} \right) \quad (6.7)$$

6.2.5 Trade-Level Focus

The framework emphasizes trade-level statistics rather than aggregate equity curves to isolate the intrinsic expectancy of signals. This avoids distortions arising from compounding effects or variable trade frequency.

6.2.6 Cross-Sectional Aggregation

Let $R^{(j)}$ denote the average trade return for asset j . The cross-sectional set:

$$\{R^{(1)}, R^{(2)}, \dots, R^{(M)}\}$$

is used to evaluate robustness across the NIFTY50 universe.

This transforms the analysis into a cross-sectional consistency test, identifying whether performance is broadly distributed or concentrated in specific assets.

6.2.7 Limitations

The framework has several constraints. It does not incorporate portfolio-level allocation or diversification effects, assumes constant slippage across all trades, and uses fixed parameters without optimization across regimes or assets. The sample period is limited, restricting long-term generalization.

Accordingly, results should be interpreted as properties of signal quality under controlled conditions rather than as direct trading performance.

6.3 Pure Indicator Strategies

This baseline layer evaluates six standard technical strategies in isolation (SMA, EMA, RSI, MACD, Bollinger Bands, and ATR breakout) Murphy 1999; Wilder 1978; Bollinger 2001; Appel 1979. The objective is to determine whether any individual indicator possesses standalone trading value when subjected to realistic execution constraints and transaction costs.

6.3.1 Empirical Findings

The empirical results provide strong and consistent evidence against exploitable intraday alpha in standalone indicators. Across all six strategies, average return per trade is negative, ranging from -0.0012 for SMA to about -0.0020 for RSI and Bollinger Bands, and profit factors remain below one.

Table 6.1 summarizes the core performance metrics across all strategies.

Trade frequency varies substantially, with ATR generating nearly 29,000 trades compared to roughly 3,500–8,000 for other indicators. This variation plays a central role in amplifying

Table 6.1: Summary of pure indicator strategy performance

Strategy	Trades	Avg Return	Win Rate	Profit Factor	Sharpe	Max DD (%)
SMA	3952	-0.0012	0.284	0.735	-0.404	-12.73
EMA	3585	-0.0013	0.253	0.716	-0.401	-12.26
RSI	3782	-0.0020	0.466	0.479	-0.888	-15.24
MACD	7871	-0.0014	0.266	0.601	-0.975	-21.58
Bollinger	3599	-0.0019	0.530	0.531	-0.776	-14.21
ATR	29236	-0.0015	0.162	0.245	-6.184	-58.92

cost drag.

A key structural feature is the divergence between win rate and profitability. Mean-reversion strategies such as RSI and Bollinger Bands achieve relatively high win rates ($p \approx 0.47$ and $p \approx 0.53$ respectively), yet remain deeply unprofitable. For RSI, the empirical decomposition is:

$$p \approx 0.466, \quad \bar{R}_{\text{win}} \approx 0.0040, \quad \bar{R}_{\text{loss}} \approx -0.0073$$

which yields:

$$\mathbb{E}[R] \approx 0.466 \cdot 0.0040 + 0.534 \cdot (-0.0073) - 0.0007 \approx -0.0020 \quad (6.8)$$

Despite winning nearly half the time, losses are almost twice as large as gains, and transaction costs further reinforce negative expectancy. Conversely, trend-following strategies such as SMA and EMA exhibit the opposite structure: average gains exceed losses, but extremely low win rates (around 25%) prevent profitability. These two failure modes—high win rate with large losses, and low win rate with insufficient frequency of gains—collectively explain the absence of positive expectancy.

Cross-Strategy Comparison The comparative results highlight that increasing structural sophistication at the indicator level does not translate into improved economic performance. The best-performing strategy by average return (SMA) still yields a negative expectancy of -0.0012 per trade, while EMA provides the least negative Sharpe ratio (-0.40). At the opposite extreme, the ATR strategy exhibits the weakest risk-adjusted performance, with a Sharpe ratio below -6 and a maximum drawdown approaching -59% .

Figure 6.1 further illustrates that all strategies occupy the negative-return region of the risk-return space. No strategy achieves a favorable trade-off between return and volatility, reinforcing the conclusion that profitability is not attainable within this class of models.

Trade Frequency and Cost Amplification A defining structural weakness of intraday indicator strategies is excessive turnover. The ATR strategy alone generates more than 29,000 trades over the sample period, implying extremely frequent entry and exit decisions. Given a

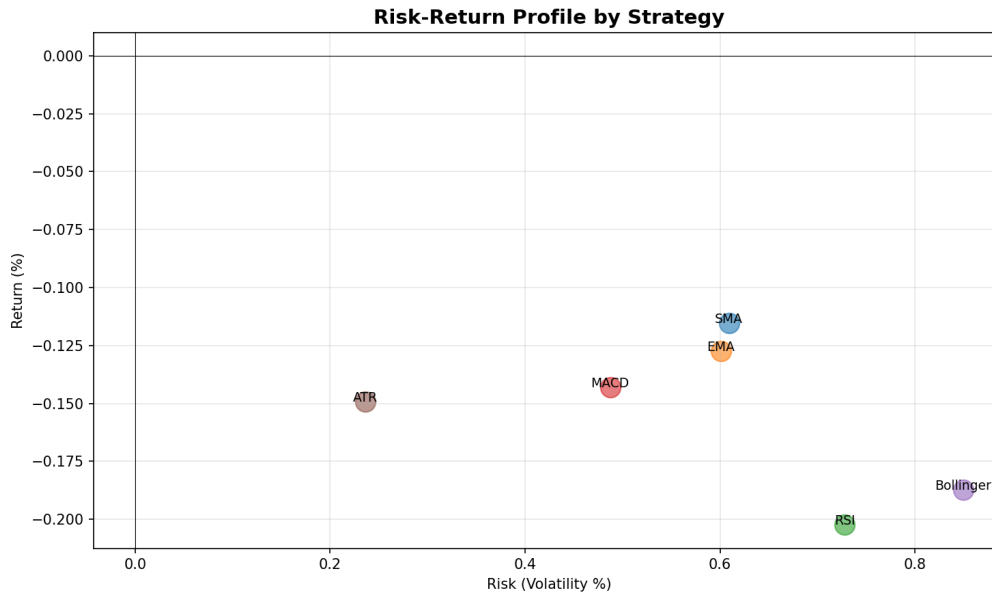


Figure 6.1: Risk-return profile of pure strategies. All strategies lie in the negative-return region, confirming absence of alpha.

fixed transaction cost of 0.07% per trade, such turnover results in substantial cumulative drag.

Even modest gross predictive power is overwhelmed under these conditions. For instance, a strategy executing approximately 20 trades per day incurs an effective daily cost of roughly 1.4%, which is orders of magnitude larger than the observed per-trade signal strength. This imbalance between signal magnitude and trading costs ensures that any weak statistical edge is eliminated in practice.

Drawdown and Tail Risk Drawdown profiles reveal persistent capital erosion across all strategies. While SMA and EMA experience relatively moderate maximum drawdowns (approximately -12%), more aggressive strategies such as MACD and ATR exhibit significantly worse tail behavior. The ATR strategy, in particular, reaches a maximum drawdown close to -59% , reflecting the compounding effect of frequent small losses.

Return distributions further exhibit negative skew, with heavier left tails across all strategies. This indicates that losses are not only frequent in some cases, but also disproportionately large when they occur, reinforcing the observed payoff asymmetry.

Cross-Sectional Heterogeneity Although aggregate results are uniformly negative, limited cross-sectional variation is observed. Certain stocks exhibit mildly positive returns under specific strategies (e.g., ONGC and SUNPHARMA under RSI, LT under SMA/EMA). However, these effects are weak, inconsistent, and not robust across strategies or assets.

Figure 6.3 shows that apparent profitability is highly localized and does not generalize across the universe, suggesting that such instances are more consistent with statistical noise than with genuine predictive edge.

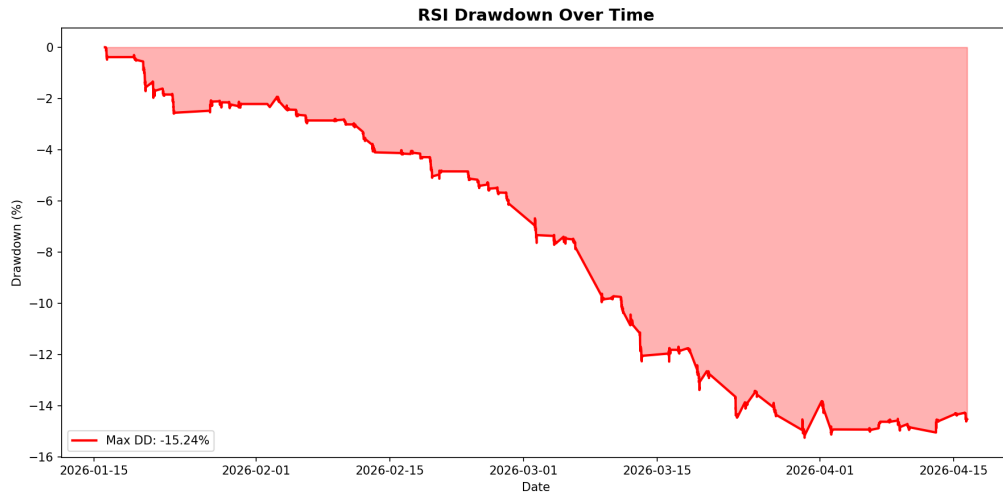


Figure 6.2: Representative drawdown curve (RSI). Persistent drawdowns indicate absence of recovery regimes.

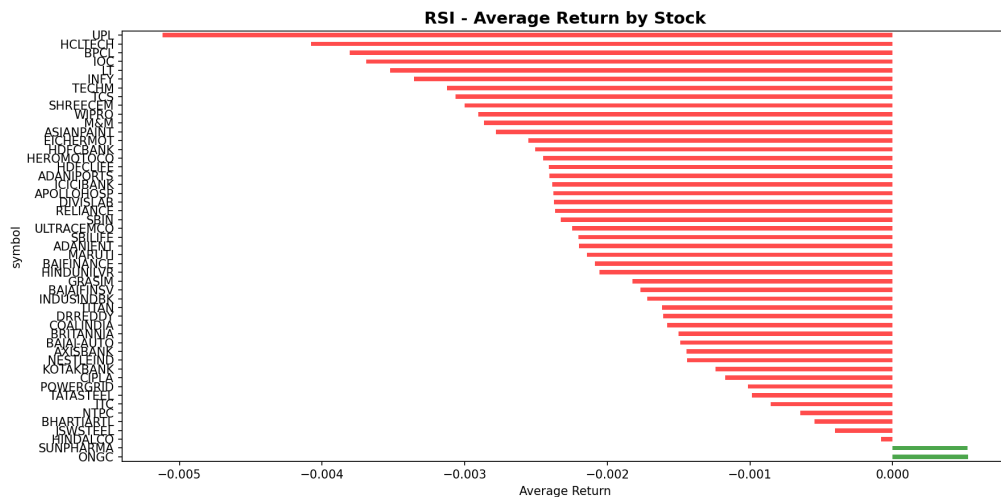


Figure 6.3: Cross-sectional performance dispersion (RSI). Limited pockets of positive returns exist but lack robustness.

6.3.2 Key Insight

The failure of pure indicator strategies arises from a structural interaction between weak predictive signals, adverse payoff asymmetry, and transaction costs. Even when indicators exhibit statistically non-random behavior, the magnitude of the signal is insufficient to overcome trading frictions. High turnover amplifies this effect, converting small per-trade inefficiencies into large cumulative losses.

6.3.3 Limitations at This Level

The results at this level are constrained by the inherent sensitivity of single indicators to intra-day noise and microstructure effects. Fixed parameter choices ignore cross-asset heterogeneity,

while next-bar execution reduces the effective strength of short-horizon signals. Most critically, the high trading frequency characteristic of these strategies amplifies cost drag, ensuring that even marginal predictive power is economically unviable.

6.4 Filtered Strategies

Filtering introduces conditional structure to trading rules by restricting signals to specific market regimes. The underlying hypothesis is that raw technical signals are contaminated by noise, and that conditioning on trend or volatility should improve signal quality by selecting higher-probability trade setups.

Five filtered variants are evaluated: RSI + Trend (price > EMA50), RSI + Low Volatility (ATR below rolling median), Bollinger Bands + Trend, Bollinger Bands + Low Volatility, and a dual-confirmation rule (RSI oversold $\cap$ Bollinger lower band).

All strategies are evaluated under identical transaction costs and slippage assumptions to ensure comparability.

6.4.1 Empirical Results

Table 6.2 reports the aggregate performance metrics across all filtered strategies.

Table 6.2: Performance summary of filtered strategies

Strategy	Trades	Avg Return	Win Rate	Profit Factor	Sharpe	Max DD (%)
RSI_Trend	684	-0.0015	46.20%	0.4608	-0.7772	-2.12
RSI_Volatility	2508	-0.0022	45.81%	0.3934	-1.0273	-10.70
BB_Trend	362	-0.0020	52.76%	0.4071	-0.6519	-1.50
BB_Volatility	1966	-0.0023	55.44%	0.3877	-0.9799	-9.05
Dual	2340	-0.0023	50.68%	0.4835	-0.8168	-10.87

Filtering has a clear and measurable effect on trade frequency. Compared to unfiltered strategies, the number of trades is reduced substantially, with the most restrictive specification (BB_Trend) generating only 362 trades, while volatility-based filters remain more permissive (e.g., 2508 trades for RSI_Volatility). This confirms that regime conditioning successfully removes a large fraction of raw signals.

However, this reduction in activity does not translate into improved profitability. Across all filtered strategies, average returns remain negative, profit factors remain well below 1.0, and Sharpe ratios remain negative.

The best-performing strategy by average return (RSI_Trend) still produces -0.0015 per trade, while the highest profit factor (Dual) reaches only 0.4835. None of the strategies approach the breakeven threshold.

6.4.2 Win Rate vs. Expectancy Decomposition

A key observation is the apparent improvement in win rates under filtering. For example, BB_Volatility achieves a win rate of 55.44%, and BB_Trend exceeds 52%. At first glance, this suggests that filtering improves signal quality.

However, this improvement is misleading when viewed in isolation. The profitability of a strategy depends on the full expectancy:

$$E[R] = p \cdot \text{AvgWin} + (1 - p) \cdot \text{AvgLoss}$$

Despite higher win rates, the magnitude of losses dominates gains across all filtered strategies. For example, BB_Volatility has Avg Win ≈ 0.0026 and Avg Loss ≈ -0.0083 , RSI_Volatility has Avg Win ≈ 0.0031 and Avg Loss ≈ -0.0066 , and Dual has Avg Win ≈ 0.0042 and Avg Loss ≈ -0.0089 .

Losses are consistently 2–3 times larger than gains. As a result, even with win rates above 50%, the expected return remains negative. Filtering improves the *probability of winning*, but not the *payoff structure*.

6.4.3 Distributional Effects

Filtering has a visible impact on the distribution of returns. As shown in Figure 6.4, the distribution becomes tighter, indicating reduced variance and fewer extreme outcomes on the positive side.

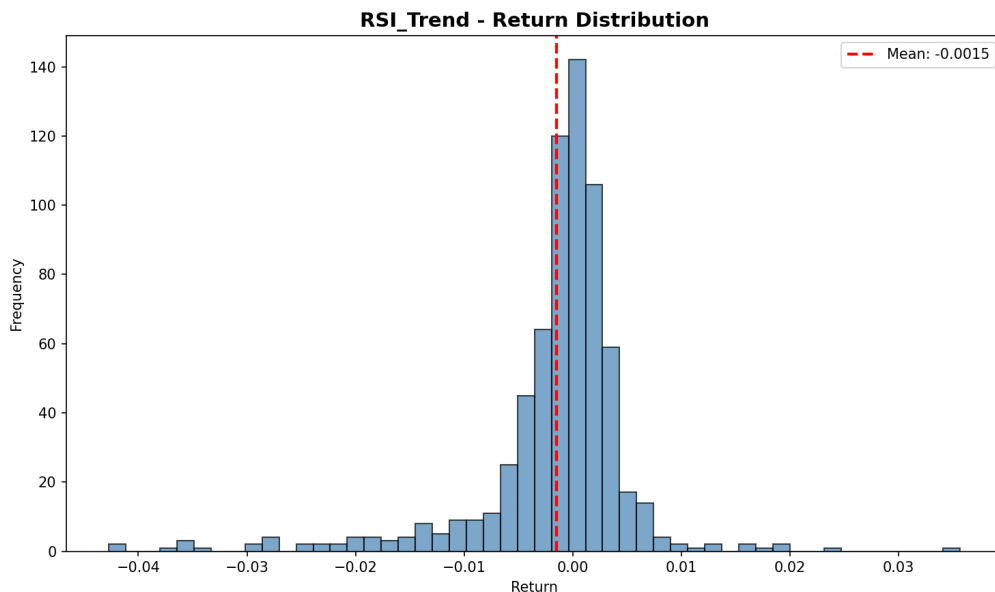


Figure 6.4: Return distribution under RSI + Trend filtering. The dispersion is reduced, but the left tail remains significant.

However, the negative tail persists across all strategies. Large adverse moves are not

eliminated by filtering, and these rare but severe losses dominate cumulative performance. This explains why profit factors remain below 1.0 even when distributions appear more concentrated.

6.4.4 Cross-Strategy Comparison

Figure 6.5 compares key performance metrics across all filtered strategies.

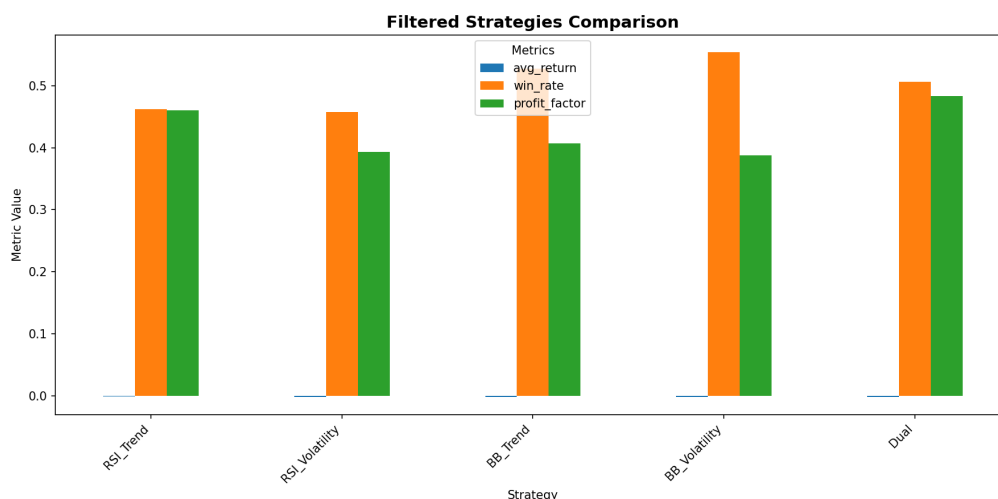


Figure 6.5: Comparison of filtered strategies across average return, win rate, and profit factor.

Two patterns emerge. Trend-based filters reduce drawdowns but also reduce opportunity, resulting in fewer trades without meaningful improvement in returns. Volatility-based filters increase trade count but amplify exposure to large losses, leading to worse drawdowns and lower Sharpe ratios.

The dual-confirmation strategy slightly improves the profit factor relative to individual filters but remains far from profitability, indicating that combining weak signals does not produce a strong signal.

6.4.5 Key Insight

Filtering improves *signal selectivity* but does not improve *signal profitability*.

The results indicate that noise reduction removes marginal trades but does not eliminate structurally unprofitable trades. Higher win rates are achieved at the cost of reduced participation, without correcting payoff asymmetry. The dominant driver of negative performance is the imbalance between average gains and losses, which filtering does not address.

In effect, filtering exposes the true nature of the underlying signals: their expected value is negative, even under favorable regimes. Conditioning cannot convert a negative-expectancy process into a positive one without altering the payoff structure.

6.4.6 Limitations of Regime Filtering

- Filtering reduces sample size, increasing statistical variance and sensitivity to outliers.
- Fixed regime definitions (e.g., EMA50, median ATR) are not adaptive to asset-specific dynamics.
- Improvements in win rate can be misleading in the presence of asymmetric return distributions.
- Filtering operates only at the selection level and does not modify trade management, entry timing, or exit efficiency.

6.5 Combined Strategies

This layer introduces multi-signal strategies that require alignment across indicators or regimes. The objective is to test whether stronger confirmation can transform weak individual signals into economically viable trading rules. Four variants are evaluated: Mean Reversion, Trend Pullback, Breakout Trend, and a Strict Combination requiring multiple simultaneous conditions.

6.5.1 Empirical Findings

Table 6.3 reports the aggregate performance across all stocks.

Table 6.3: Performance of Combined Strategies

Strategy	Trades	Win Rate	Profit Factor	Avg Return	Avg Win	Avg Loss
Mean Reversion	2674	49.44%	0.5145	-0.00204	0.00436	-0.00829
Trend Pullback	1249	53.56%	0.5282	-0.00164	0.00343	-0.00750
Breakout Trend	7037	16.60%	0.5616	-0.00149	0.01150	-0.00408
Strict Combo	111	54.05%	0.4393	-0.00207	0.00300	-0.00802

Several structural improvements are evident relative to both pure and filtered strategies. Profit factors increase modestly (up to ≈ 0.56), some strategies achieve higher win rates (Trend Pullback and Strict Combo near 54%), and breakout strategies exhibit strong payoff asymmetry with average wins ($\approx 1.15\%$) nearly three times larger than average losses.

Despite these improvements, all strategies continue to exhibit negative average returns and profit factors below 1.0. The magnitude of improvement is insufficient to overcome transaction costs and residual noise.

The Breakout Trend strategy is particularly instructive. It exhibits the strongest structural edge but still fails economically. Its expectancy can be expressed as:

$$\mathbb{E}[R] = p \cdot \bar{R}_{\text{win}} + (1 - p) \cdot \bar{R}_{\text{loss}} - \text{costs} \quad (6.9)$$

With empirical values:

$$p \approx 0.166, \quad \bar{R}_{\text{win}} \approx 0.0115, \quad \bar{R}_{\text{loss}} \approx -0.0041$$

Even with favorable payoff asymmetry, the low win probability implies that losses dominate in frequency. After accounting for trading costs, the net expectancy remains negative. This demonstrates a clear separation between *statistical edge* and *economic profitability*.

Figure 6.6 summarizes cross-strategy performance, while Figure 6.7 highlights the skewed but still loss-making return distribution of the breakout strategy.

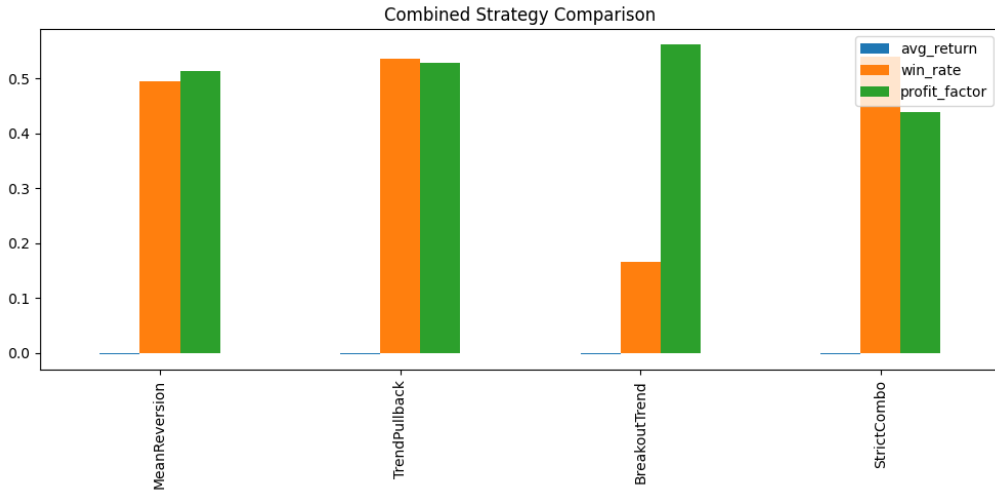


Figure 6.6: Combined strategies improve structural metrics (win rate, payoff asymmetry), but none achieve positive expectancy.

6.5.2 Key Insight

Combining signals improves *structure* but not *magnitude*. The strategies become more interpretable and exhibit desirable properties such as trend alignment or asymmetric payoffs, but the underlying alpha remains too weak.

This section establishes a critical result: even when signals are combined to enforce stronger confirmation, the resulting strategies fail to achieve positive expectancy. The limiting factor is not signal design complexity, but the absolute scale of predictive edge relative to costs.

6.5.3 Limitations at This Level

- Multi-signal confirmation reduces trade frequency (e.g., Strict Combo: only 111 trades), increasing variance and reducing statistical reliability.
- Improvements in payoff asymmetry do not compensate for low win probabilities in breakout-style strategies.

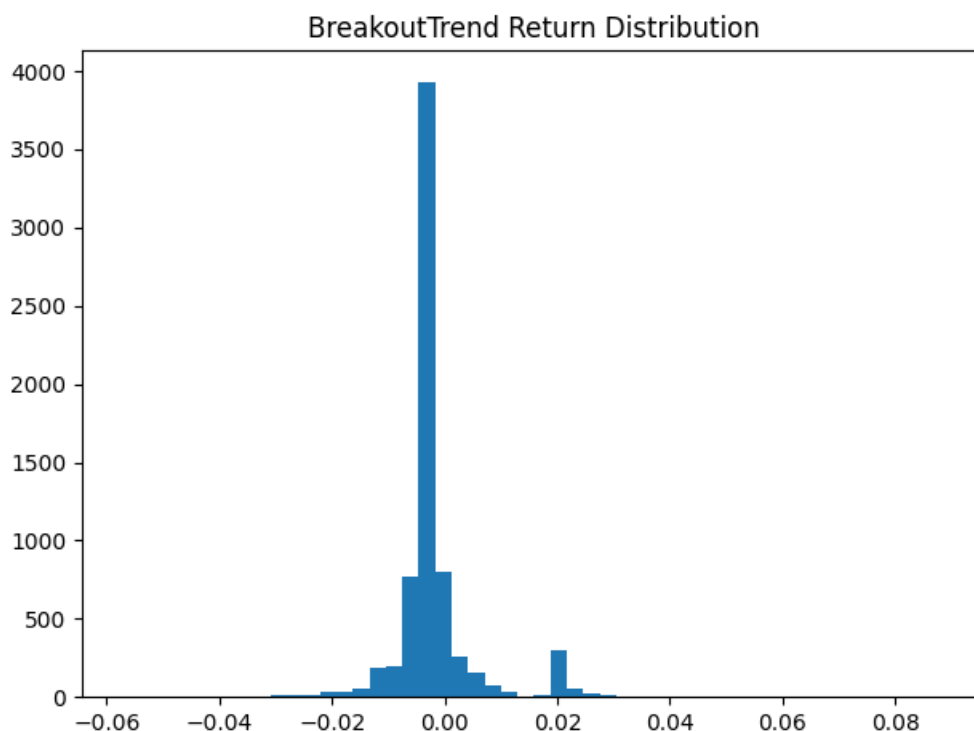


Figure 6.7: Breakout Trend exhibits strong positive skew (large wins, frequent small losses), yet remains unprofitable after costs.

- Transaction costs remain a dominant factor, especially for high-frequency strategies such as Breakout Trend (7000+ trades).
- Strategy parameters are fixed globally and do not adapt to asset-specific behavior, limiting potential gains.

6.6 Stock-Level Analysis

Stock-level analysis evaluates cross-sectional robustness using the dual-confirmation combined strategy (RSI + Bollinger Bands). This is the most stringent test of whether any apparent edge generalizes across assets rather than being driven by aggregate effects.

6.6.1 Empirical Findings

The results provide strong evidence of cross-sectional failure. Only 2 out of 49 stocks (4.1%) are profitable, and only 2 exhibit positive Sharpe ratios. The average return across stocks is -0.002247 , with a median of -0.002273 , indicating that losses are not driven by a few outliers but are pervasive across the universe.

Despite a mean win rate above 50%, profitability remains negative. This directly confirms the loss-dominance mechanism observed earlier: gains are smaller than losses in magnitude.

Table 6.4: Stock-Level Summary Statistics

Metric	Value
Total stocks analyzed	49
Profitable stocks	2 (4.1%)
Positive Sharpe ratio	2 (4.1%)
Mean return	-0.002247
Median return	-0.002273
Return standard deviation	0.001341
Best stock return	0.001475
Worst stock return	-0.005522
Mean win rate	50.76%
Mean profit factor	0.5263
Mean Sharpe ratio	-3.4057
Mean max drawdown	-13.49%
Worst max drawdown	-25.19%

Empirically,

$$\bar{R}_{\text{win}} \approx 0.0042 \quad \text{vs.} \quad \bar{R}_{\text{loss}} \approx -0.0089 \quad (6.10)$$

Thus, even with frequent winning trades, the larger losses dominate overall returns.

Table 6.5: Top and Bottom Performing Stocks (by Average Return)

Symbol	Avg Return	Win Rate	Profit Factor	Sharpe
ONGC	0.001475	60.5%	1.61	2.60
SUNPHARMA	0.000015	65.1%	1.01	0.03
TECHM	-0.005522	43.2%	0.31	-5.42
UPL	-0.005009	51.1%	0.34	-4.82
INFY	-0.004510	31.3%	0.23	-6.31

Cross-sectional dispersion is wide but asymmetrically negative. While a small number of stocks (e.g., ONGC) show strong performance, the majority exhibit consistently poor returns with deeply negative Sharpe ratios. Importantly, even among the best-performing stocks, only one achieves a clearly meaningful profit factor above 1.

Figure 6.8 shows that the entire return distribution is shifted below zero. Figure 6.9 demonstrates that high win rates do not translate into profitability. Figure 6.10 confirms that risk-adjusted performance is overwhelmingly negative across the cross-section.

6.6.2 Key Insight

Cross-sectional instability provides decisive evidence against the existence of a robust trading edge. If a strategy were genuinely predictive, performance would cluster around a positive

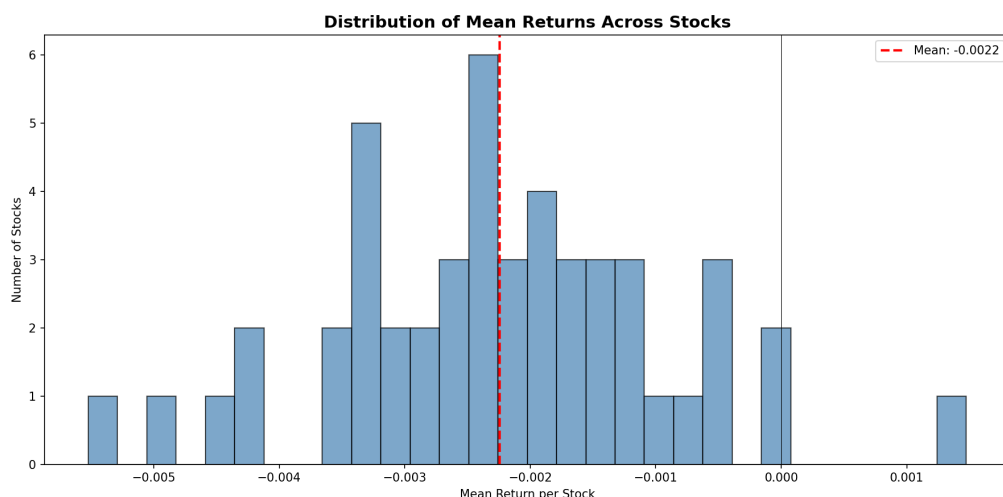


Figure 6.8: Return distribution across stocks is centered below zero, indicating systematic unprofitability.

mean across assets. Instead, the distribution is centered on negative returns with only 4% of stocks profitable.

This proportion is consistent with statistical noise. Under a null hypothesis of no edge, a small fraction of assets (on the order of 5%) will appear profitable purely by chance. The observed results align closely with this expectation, indicating that the few profitable cases are likely idiosyncratic rather than systematic.

The strongly negative mean Sharpe ratio (-3.4) further indicates that the strategy is not merely weak but structurally detrimental when applied across assets.

6.6.3 Limitations at This Level

- Trade counts per stock are relatively low (mean ~ 48), which can introduce estimation noise in individual stock metrics.
- A small number of outliers (e.g., ONGC) may reflect favorable microstructure conditions rather than transferable signal strength.
- No portfolio construction or capital allocation scheme is applied; diversification effects are not captured.
- The small fraction of profitable stocks is statistically consistent with randomness and does not imply genuine predictive power.

6.7 Comparative Discussion

The three strategy layers show a clear progression in sophistication but not in profitability. Pure strategies are noisy but simple; filtered strategies reduce noise but constrain opportunity;

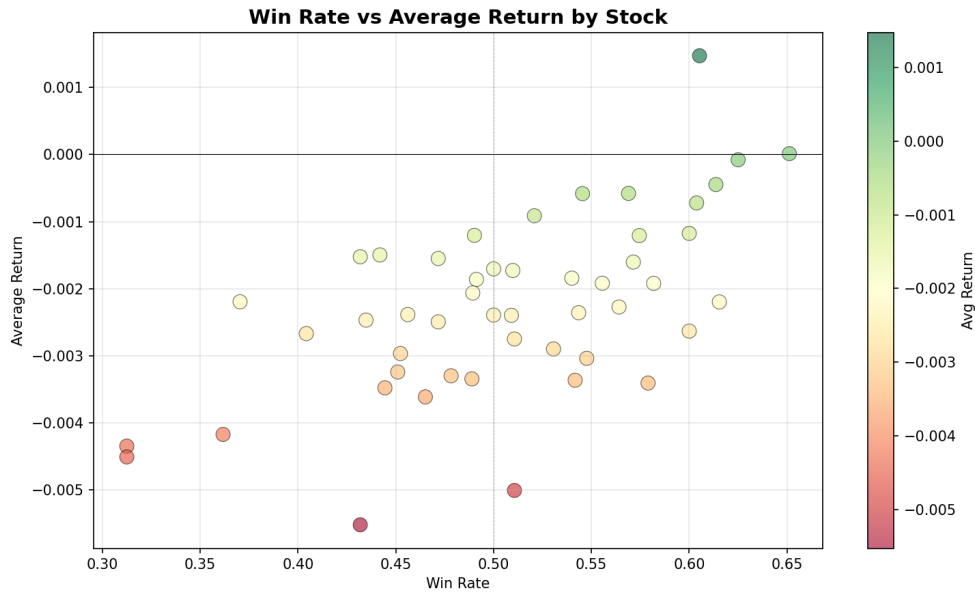


Figure 6.9: Stocks with win rates above 50% still exhibit negative returns, confirming loss dominance.

combined strategies improve structure but reduce frequency. Yet across all levels, profit factors remain below 1.0 and average trade returns are negative.

This monotone failure pattern is instructive. It indicates that the obstacles to profitability are not superficial (e.g., "poor indicator choice") but structural. Observed failures stem from four core mechanisms:

- **Cost drag:** Transaction costs (0.07% per trade) consume gross alpha. At 20 trades per strategy, cumulative drag is 1.4%—a large haircut to thin returns.
- **Loss asymmetry:** Average losses significantly exceed average wins across all strategies and stocks. This dominates the expectancy calculation even before costs are applied.
- **Frequency-profitability tradeoff:** Reducing trade frequency (filtering, combination) lowers cost drag per signal but reduces opportunities to accumulate marginal edges. With negative gross alpha, volume cannot save the strategy.
- **Cross-sectional instability:** Performance is concentrated in a few outlier stocks. No strategy generalizes across assets, indicating lack of robust edge.

None of these mechanisms are addressable through indicator refinement alone. Each is a fundamental constraint of the signal-cost-payoff ecosystem.

6.8 Synthesis and Link to Signal-Level Results

A key objective of this analysis is to evaluate whether predictive patterns observed at the signal level translate into economic value. Chapter 4 demonstrated that several indicators

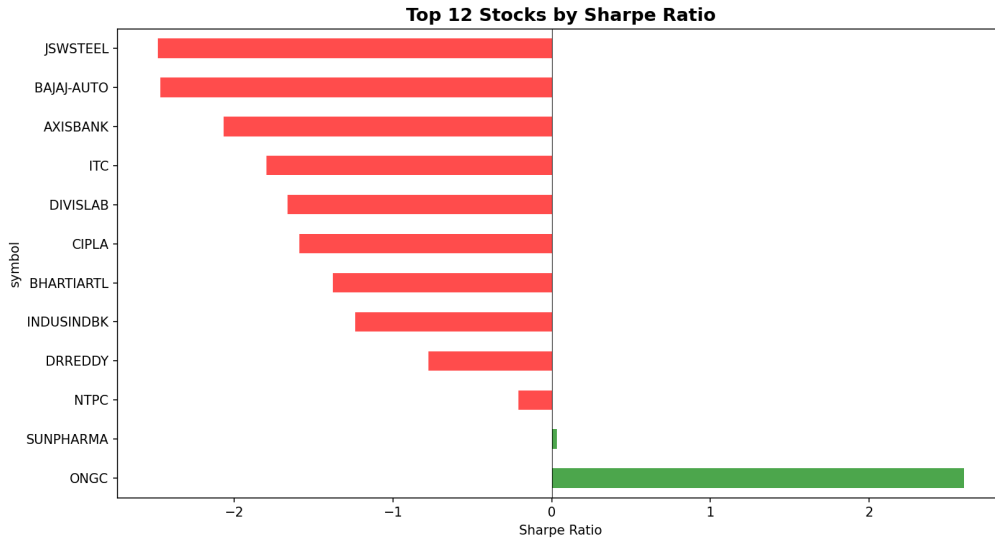


Figure 6.10: Sharpe ratios are predominantly negative, indicating poor risk-adjusted performance across nearly all stocks.

exhibited non-random behavior at the signal level, with statistically significant effects on forward returns. Yet strategy-level analysis reveals that this predictability fails to survive implementation.

The gap between signal-level predictability and strategy-level profitability arises from three mechanisms: transaction costs (even 2–3 bps signals are consumed by 7 bps in fees and slippage), execution lag (next-bar execution degrades short-horizon signals), and payoff asymmetry (losses are larger than gains even when signals are directionally correct).

These factors combine multiplicatively. A signal with modest predictive power ($p = 0.51$, indicating 51% directional accuracy) that faces adverse payoff asymmetry (losses twice as large as wins) and costs will produce negative expectancy:

$$\mathbb{E}[R] \approx 0.51 \times 0.003 + 0.49 \times (-0.006) - 0.0007 < 0 \quad (6.11)$$

This demonstrates why statistical significance at the signal level does not imply trading profitability.

6.8.1 Core Empirical Pattern

Across all strategy classes and levels, a recurring pattern dominates: win rates are moderate (40–55%), average losses exceed average gains by 50–100%, profit factors remain below 1.0, and Sharpe ratios are uniformly negative.

This pattern does not emerge from aggregation artifacts or outlier-driven results. It holds at the individual-stock level, at the strategy level, and across all three strategic sophistication tiers. The universality of the pattern is evidence that it reflects fundamental market properties rather than specific indicator failures.

Strategies frequently win often but lose more, resulting in systematically negative returns despite moderate win rates.

6.8.2 Implications for Strategy Design

The results establish that improving signal quality alone is insufficient for profitability. Neither filtering nor signal combination addresses the underlying expectancy deficit. Position sizing and risk management cannot fix a negative-expectancy signal, and cost reduction is necessary but not sufficient at observed signal strengths. High-frequency trading amplifies cost drag, cross-sectional robustness remains a hard requirement, and tail-risk management matters more than boosting win rates.

6.9 Conclusion

Technical indicator-based strategies fail to deliver robust profitability in the intraday NIFTY50 setting, despite evidence of non-random signal behavior at the signal level. The limiting factors are structural: transaction costs, execution lag, and adverse payoff asymmetry.

Across pure, filtered, and combined strategies, profit factors remain below 1.0, average returns are negative, and risk-adjusted performance is poor across all 49 stocks. Increased structural sophistication improves signal structure but does not close the profitability gap.

Two findings are particularly stark:

1. **The win-rate paradox:** Many stocks and strategies exhibit win rates above 50%, yet remain unprofitable. This demonstrates that losses are structurally larger than gains, a property indicator design alone cannot fix.
2. **Cross-sectional instability:** Only 4% of stocks are profitable when trading the best combined strategy. This lack of generalization rules out any claim of systematic edge; the apparent profitability in a few stocks is consistent with noise.

In aggregate, the evidence indicates that technical indicator signals, though statistically detectable, do not provide an economic edge sufficient to overcome realistic trading costs and execution frictions. Practitioners seeking to exploit intraday technical patterns face a fundamental profitability barrier that extends beyond the current indicator set or parameter choices.

Chapter 7

Conclusion & Future Work

7.1 Summary

This study evaluated whether common technical indicators provide predictive value at 5-minute horizons and whether that signal can be translated into profitable intraday strategies in the NIFTY50 universe. Using 49 stocks over roughly 60 trading days, signal-level analysis showed that pooled forward returns are economically negligible and any nominal significance disappears after multiple-testing control. Strategy backtests reinforce the same conclusion: after realistic costs and execution lag, most indicator-based strategies exhibit negative average returns, profit factors below 1.0, and negative risk-adjusted performance.

Single-asset experiments highlight the strongest baseline as the Opening Range Breakout (ORB), and VWAP works best as a confirmation filter rather than a standalone signal. However, these improvements do not yield consistent profitability across assets. The multi-asset analysis makes the limitation explicit: even the best combined strategy is profitable for only a small fraction of stocks, and a win rate above 50% is insufficient when losses are structurally larger than gains. Overall, the evidence indicates that standard technical indicators may contain weak statistical structure but do not provide a tradable edge at intraday horizons once costs and payoff asymmetry are considered.

7.2 Contributions

- A unified evaluation pipeline linking signal-level predictability to strategy-level backtesting under realistic intraday constraints.
- Cross-asset pooled testing with FDR correction to assess the robustness of indicator signals at multiple horizons.
- Comparative evidence across pure, filtered, and combined strategies, showing that added indicator complexity does not overcome structural profitability limits.

- Empirical documentation of the win-rate paradox and cross-sectional instability in NIFTY50 intraday technical strategies.

7.3 Future Work

- Regime detection and volatility targeting to condition signals on market state.
- More realistic execution models with latency, partial fills, and order book impact.
- Portfolio-level construction and capital allocation to test whether diversification can improve expectancy.
- Dynamic position sizing and risk controls that respond to intraday liquidity and volatility.
- Longer and more diverse samples to assess stability across market regimes.

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Appendix A

Additional Material

A.1 Additional Figures

A.2 Extended Tables

A.3 Implementation Details